AN EXPLICIT FORMULA FOR THE LODAY ASSEMBLY

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ABSTRACT. We give an explicit description of the Loday assembly map on homotopy groups when restricted to a certain subgroup coming from the Atiyah-Hirzebruch spectral sequence. This proves and generalises a formula about the Loday assembly map on the first homotopy group written down in the survey article by Lück and Reich. Furthermore, we use this new result to prove the Loday assembly for the integral group ring of cyclic group of order 2 is a bijection on the second homotopy group, and is an injection when one considers n-fold direct sum of cyclic group of order 2. Finally, we show that our formula can be used to compute the universal assembly on homotopy.

CONTENTS

1. Introduction

Classically, algebraic K-theory deals with following three Abelian groups $K_0(R)$, $K_1(R)$, $K_2(R)$ associated to a ring R. See [Ros95] for their definitions. For a sufficiently nice space X, topologists have found geometric applications of the groups $K_i(\mathbb{Z}[\pi_1(X)])$ for $i = 0, 1, 2$, where $\mathbb{Z}[\pi_1(X)]$ is the integral group ring of the fundamental group of the space X. For example, an important application of K_0 was found by Wall [Wal65]. He defined an element $\chi(X)$ in a quotient group $\mathrm{Wh}_0(\pi_1(X))$ of $K_0(\mathbb{Z}[\pi_1(X)])$, the so-called *Wall's finiteness obstruction*, and showed that X is homotopy equivalent to a finite CW-complex if and only if $\chi(X) = 0$. The second example, which is due to Whitehead, is an attempt to classify manifolds of dimension at least five. In a collection of work [Whi39; Whi41; Whi50], he defined the *Whitehead torsion* $\tau(f) \in \mathrm{Wh}_1(\pi_1(X))$ of a continuous map f, where $\mathrm{Wh}_1(\pi_1(X))$ is a quotient of $K_1(\mathbb{Z}[\pi_1(X)])$. In particular, the vanishing of the Whitehead torsion allowed Smale to prove the Poincaré Conjecture in dimensions greater than four [Sma61].

For an ideal $I \subseteq R$, there is a natural exact sequence

$$K_2(R) \to K_2(R/I) \to K_1(R,I) \to K_1(R) \to K_1(R/I) \to K_0(R,I) \to K_0(R) \to K_0(R/I) \tag{1.0.1}$$

(see [Ros95, Theorem 4.3.1 on page 200]), and a lot of effort went into the search for definitions for $K_i(R)$ with $i \geq 3$ to extend this exact sequence to the left. The first widely accepted definition was due to Quillen [Qui72; Qui73], for which he was awarded the Fields Medal in 1978. His idea was that one should try to construct the groups $K_i(R)$ not one at a time but all at once, as homotopy groups of a certain topological space $K(R)$

$$K_i(R) = \pi_i(K(R)),$$

constructed to reflect structures of the category of finitely generated projective R-modules. It was later discovered by Gersten and Wagoner that the space $K(R)$ is an infinite loop space [Ger72; Wag72], and hence an Ω-spectrum, making algebra K-theory part of stable homotopy theory. Several definitions of algebraic K-theory came out to account this property, notably Waldhausen's $S_\bullet$-construction [Wal78c]. Nowadays, people understand algebraic K-theory as a machine that takes in nice data to produce Ω-spectra, or more precisely, a functor

$$\mathbb{K} : \mathcal{S}\mathrm{ym}\mathcal{M}\mathrm{on}\mathcal{C}\mathrm{at} \to \Omega\text{-}\mathcal{S}\mathrm{pectra} \tag{1.0.2}$$

from symmetric monoidal categories to Ω-spectra such that the homotopy groups

$$\pi_i\left(\mathbb{K}\left(\mathcal{P}\mathrm{roj}_R^{\mathrm{fg}}\right)\right)$$

of the Ω-spectrum obtained by evaluating at the category $\mathcal{P}\mathrm{roj}_R^{\mathrm{fg}}$ of finitely generated projective R-modules recovers the classical K-groups of R for $i = 0, 1, 2$ and the higher K-groups defined by Quillen.

However, the general consensus is that computing homotopy groups is difficult. Hence, investigating algebraic K-theory of rings is not an easy task. On the other hand, homology is more accessible—there are excisions, the Mayer-Vietoris sequence, and even spectral sequences in homology that are more user-friendly than they would be in homotopy. This is the story of assembly—to approximate homotopy theory by a generalised homology theory.

1.1. **History and Motivations.** In his dissertation [Lod76], Loday defined a map

$$\alpha_{\mathrm{Loday}} : BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}} \to \mathbb{K}_{R[G]}^{\mathrm{GW}}$$

of spectra, which is now known as the Loday assembly (see Definition 3.3), to unify the classical Whitehead groups $\mathrm{Wh}_i(G)$ for $i = 0, 1, 2$ studied by Wall [Wal65], Whitehead [Whi39; Whi41; Whi50], and Hatcher-Wagoner [HW73]. Before this work appeared, Wall defined a version of assembly

$$A_i : H_i(BG; \mathbb{Q}) \to L_i(\mathbb{Z}[G]) \otimes \mathbb{Q}$$

for L-theory. Furthermore, he showed that the injectivity of A_i implies the classical Novikov Conjecture about homotopy invariance of the higher signature of a closed, oriented manifold M when $G = \pi_1(M)$ [Wal70, 17H]. The converse is also true [KM81], and therefore the classical Novikov Conjecture is equivalent to the injectivity of the map A_i for all i. Motivated by this, Hsiang cast the following conjecture in his 1984 ICM address:

Conjecture 1.1 (K-theoretic Novikov Conjecture, [Hsi84])
Let R be a regular ring, and G be a torsion-free group. Then the map

$$\pi_i(\alpha_{\mathrm{Loday}}) \otimes \mathrm{id}_{\mathbb{Q}} : \pi_i(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \otimes \mathbb{Q} \to K_i(R[G]) \otimes \mathbb{Q}$$

is injective for all i.

Attempts on answering the K-theoretic Novikov Conjecture 1.1 often leads to the creation of new mathematics, most notably the Topological Cyclic Homology, which allowed Bökstedt-Hsiang-Madsen to prove the following:

Theorem 1.2 ([BHM93, Theorem 9.13 on page 535])
The K-theoretic Novikov Conjecture 1.1 is true if G is homologically finite.

A stronger statement is also possible:

Conjecture 1.3 (Classical Farrell-Jones Conjecture)
Let R be a regular ring, and G be a torsion-free group. Then the map

$$\pi_i(\alpha_{\mathrm{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \to K_i(R[G])$$

is an isomorphism for all i.

This conjecture can be understood as a conceptual approach to computing algebraic K-theory of a group ring via a homology theory, but it has important implications for a range of topics, notably the Borel Conjecture concerning the topological rigidity of closed spherical manifolds, and the Kaplansky Conjecture about the idempotents in a group ring.

The following case has been verified:

Theorem 1.4 ([BLR08, Theorem 1.1 (i) on page 58])
The Classical Farrell-Jones Conjecture 1.3 is true if the group G is word-hyperbolic.

The work presented here is motivated by the Classical Farrell-Jones Conjecture 1.3.

1.2. **What Are We Trying to Do?** The purpose of this article is to help the author to obtain his doctoral degree by studying the Loday assembly on a certain subgroup coming from the Atiyah-Hirzebruch spectral sequence for its source.

More precisely, the source of the Loday assembly

$$\alpha_{\text{Loday}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}}$$

represents a generalised homology theory. Hence, its homotopy groups can be computed by the Atiyah-Hirzebruch spectral sequence

$$\begin{aligned} E_{p,q}^2 &\cong H_p(BG_+; K_q(R)) \\ &\Rightarrow \pi_{p+q}(R[G]). \end{aligned}$$

From the construction of the spectral sequence, we know there is a subgroup $E_{1,i}^{\infty} \leq \pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right)$ coming from the 1-skeleton of the classifying space BG. We provide a formula for the restriction $\pi_{i+1}(\alpha_{\text{Loday}})|_{E_{1,i}^{\infty}}$ (see Theorem 3.8). The derivation of our formula involves extending the original Loday pairing

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

to the full K-theory space

$$\gamma'_{\text{Loday}} : \left[K_0(R) \times BGL(R)^+\right] \wedge \left[K_0(S) \times BGL(S)^+\right] \to K_0(R \otimes S) \times BGL(R \otimes S)^+$$

in a non-obvious way (see Definition 2.5). In particular,

(i) we relate the product map

$$\star' : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S)$$

by the extend pairing γ_{Loday} to the classical product maps defined by Milnor in [Mil72] (see Theorem 2.9).

(ii) We recover (and generalise) a formula written down in the survey article [LR05] by Lück-Riech about the Loday assembly on π_1 (see Corollary 3.9).

(iii) We obtain injectivity/bijectivity results on π_2 for the group ring $\mathbb{Z}[C_2^{\oplus n}]$ (see Example 4.5 and Example 4.10).

(iv) We prove that the non-injectivity phenomenon of the Loday assembly on π_2 is related to the word problem for the Steinberg group (see Theorem 4.7).

(v) We show that our formula can be used to compute the universal assembly (see Theorem 5.50).

1.3. **Organisation.** This article consists of five parts, including the introduction. In Chapter 2, we review the construction of the original Loday pairing map and extend it to the full K-theory space using the Gersten-Wagoner delooping. We then relate the induced product maps on the K-groups with the classical product maps defined by Milnor and use the extended Loday pairing to construct the non-connective Gersten-Wagoner Algebraic K-theory spectrum.

In Chapter 3, we use the extended Loday pairing to define the Loday assembly and identify the subgroup of the source that we are interested in by studying the Atiyah-Hirzebruch spectral sequence. It also contains the statement and the proof of our main result.

Chapter 4 deals with the second K-group of integral group rings. We study the extension problem for the second homotopy group of the source of the Loday assembly coming from the Atiyah-Hirzebruch spectral sequence. Then we provide injectivity/bijectivity results for the Loday assembly for n-fold direct sum of the cyclic group of order 2. We also show that non-injectivity phenomenon of the Loday assembly on π_2 is related to the word problem for the Steinberg group.

Chapter 5 is a quick summary of Sperber's work on proving the Loday assembly is the universal assembly [Spe04]. We show that our version of the Loday assembly can be used to compute the universal assembly on homotopy even with the presence of the non-obvious choice of extending the original Loday pairing.

1.4. Notations and Conventions.

1.4.1. *Rings and Modules.* All *rings* in this article are associative, unital, but not necessarily commutative. A *module* over a ring will always be understood as left module.

Given a ring R, we define its *cone* to be the ring $\mathrm{cone}\,(R)$ of locally finite matrices over R, that is,

$$\mathrm{cone}\,(R) := \left\{ A \in M(R) \;\middle|\; \begin{array}{c}\text{Each column and each row of } A\\ \text{only have finitely many non-zero entries}\end{array} \right\}. \tag{1.4.1}$$

There is an ideal $m(R)$ of $\mathrm{cone}\,(R)$ consists of infinite matrices with only finitely many non-zero entries. The *suspension of* R is defined to be the quotient

$$\Sigma R := \frac{\mathrm{cone}\,(R)}{m(R)}. \tag{1.4.2}$$

We also define

$$\Sigma^0 R := R \tag{1.4.3}$$

and inductively

$$\Sigma^n R := \Sigma\left(\Sigma^{n-1} R\right). \tag{1.4.4}$$

Note that for any ring R, we have

$$\Sigma R \cong \Sigma\mathbb{Z} \otimes R.$$

Therefore, for any two rings R, S, we have

$$\Sigma R \otimes \Sigma S \cong \Sigma^2 (R \otimes S). \tag{1.4.5}$$

1.4.2. *Topological Spaces and Spectra.* The term *topological space* (or simply *space*) will always mean a pointed, compactly generated, weak Hausdorff topological space. This includes all finite CW-complexes (and hence all topological manifolds). Maps between spaces are always pointed and continuous.

A map $f : X \to Y$ between spaces is said to satisfy some property *up to weak homotopy* if it satisfies the underlying property up to homotopy when restricted onto any compact subset. For example, a map $f : X \to Y$ is *well-defined up to weak homotopy* if the restriction $f|_K : K \to Y$ is well-defined up to homotopy for any compact subset $K \subseteq X$. Any map of spaces that is well-defined up to weak homotopy induces well-defined group homomorphisms on homotopy. We say two maps $f, g : X \to Y$ of spaces are *weakly homotopic* if the restrictions $f|_K$, $g|_K$ are homotopic for any compact subset $K \subseteq X$. Weakly homotopic maps induce the same group homomorphisms on homotopy.

By *spectrum*, we mean a sequence $\mathbb{E} = \{\mathbb{E}_i \mid i \in I\}$ of spaces, where $I = \mathbb{N} \cup \{0\}$ or $\mathbb{Z}$, together with map

$$f_i : \mathbb{S}^1 \wedge \mathbb{E}_i \to \mathbb{E}_{i+1}$$

for each $i \in I$, called the *structure maps*. An Ω-*spectrum* is a spectrum whose adjoints

$$\overline{f}_i : \mathbb{E}_i \to \Omega\mathbb{E}_{i+1}$$

of the structure maps are all homotopy equivalence. A *morphism* or a *map* (resp. *weak morphism* or a *weak map*) $\varphi : \mathbb{E} \to \mathbb{F}$ of spectra is a collection $\{\varphi_i : \mathbb{E}_i \to \mathbb{F}_i \mid i \in I\}$ of maps that commutes with the structure maps (resp. up to weak homotopy).

If $\mathbb{E}$, $\mathbb{F}$, $\mathbb{G}$ are spectra, then a pairing (resp. weak pairing) $\mu : \mathbb{E} \wedge \mathbb{F} \to \mathbb{G}$ of spectra is a collection of maps $\mu_{m,n} : \mathbb{E}_m \wedge \mathbb{F}_n \to \mathbb{G}_{m+n}$ that commute with the structure maps up to homotopy (resp. weak homotopy). We abuse notation here to use the smash product in writing down a pairing of spectra.

2. Loday Pairing

2.1. **The Loday Pairing.** Let R, S be rings, and fix isomorphisms

$$\xi_{m,n} : R^m \otimes S^n \to (R \otimes S)^{mn}\,. \tag{2.1.1}$$

In this setting, the tensor product of matrices gives a group homomorphism

$$GL\,(m, R) \times GL\,(n, S) \to GL\,(mn, R \otimes S)\,,$$

sending elementary matrices to elementary matrices. Hence we have an induced map

$$f^{R,S}_{m,n} : BGL(m, R)^+ \times BGL(n, S)^+ \to BGL(mn, R \otimes S)^+\,. \tag{2.1.2}$$

Write

$$i_{mn} : BGL(mn, R \otimes S)^+ \to BGL(R \otimes S)^+$$

to be map induced by the canonical inclusion

$$GL\,(mn, R \otimes S) \to GL\,(R \otimes S)\,.$$

We then define

Definition 2.1 (The map $\gamma_{m,n}$, [Lod76, page 332])
The map

$$\gamma_{m,n} : BGL(m, R)^+ \times BGL(n, S)^+ \to BGL(R \otimes S)^+ \tag{2.1.3}$$

is defined by the formula:

$$\gamma_{m,n}(x, y) := i_{mn} \circ f^{R,S}_{m,n}(x, y) - i_{mn} \circ f^{R,S}_{m,n}(x, y_0) - i_{mn} \circ f^{R,S}_{m,n}(x_0, y)\,. \tag{2.1.4}$$

Here, the minus sign on the RHS comes from the H-group structure of $BGL(R \otimes S)^+$; and x_0 (resp. y_0) is the base-point in $BGL(m, R)^+$ (resp. $BGL(n, S)^+$). (I.e., represented by the $m \times m$ identity matrix.)

The choice of the isomorphisms $\xi_{m,n}$ says the map $\gamma_{m,n}$ is well-defined up to **weak-homotopy**. More precisely, the map $\gamma_{m,n}$ is well-defined up to homotopy when restricted onto compact subsets of $BGL(m,R)^+ \times BGL(n,S)^+$.

One gets a map

$$\gamma : BGL(R)^+ \times BGL(S)^+ \to BGL(R \otimes S)^+ \tag{2.1.5}$$

by letting $m, n \to \infty$. This map, again, is well-defined up to weak-homotopy. Moreover, it is homotopy trivial on the wedge product $BGL(R)^+ \vee BGL(S)^+$.

Definition 2.2 (The Loday pairing map γ_{Loday}, [Lod76, Section 2.1.7 on page 333])
The map

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+ \tag{2.1.6}$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{tikzcd} BGL(R)^+ \times BGL(S)^+ \arrow[r, "\gamma"] \arrow[d, "\text{proj}"'] & BGL(R \otimes S)^+ \\ BGL(R)^+ \wedge BGL(S)^+ . \arrow[ur, "\gamma_{\text{Loday}}"'] & \end{tikzcd} \tag{2.1.7}$$

This allows us to define a multiplicative structure on algebraic K-theory.

Definition 2.3 (Loday product $\star$, [Lod76, Section 2.1.10 on page 335])
For each integers $i, j \geq 1$, we define the product

$$\star : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S) \tag{2.1.8}$$

by

$$[f] \star [g] := [\gamma_{\text{Loday}} \circ (f \wedge g)] , \tag{2.1.9}$$

where f and g are spheroids

$$f : \mathbb{S}^i \to BGL(R)^+ ,$$
$$g : \mathbb{S}^j \to BGL(S)^+ .$$

Proposition 2.4 (Properties of the Loday Product $\star$)
The product map $\star$ is

(i) natural in R and S, associative and bilinear [Lod76, Theorem 2.1.11 on page 335];

(ii) graded commutative:

$$[f] \star [g] = (-1)^{ij} ([g] \star [f]) \tag{2.1.10}$$

whenever $[f] \in K_i(R)$ and $[g] \in K_j(S)$ [Lod76, Theorem 2.1.12 on page 335];

(iii) the product $\{u\} \star \{v\}$ is the inverse of the Steinberg symbol:

$$\{u\} \star \{v\} = -\{u, v\}_{\mathrm{St}} \tag{2.1.11}$$

when $i = j = 1$ [Lod76, Proposition 2.2.3 on page 337]. Here, we write the Abelian group $K_2(R \otimes S)$ additively.

2.2. **The Extended Loday Pairing.** We want to modify the Loday pairing map

$$\gamma_{\mathrm{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

to a map

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+,$$

where

$$K_R^+ := K_0(R) \times BGL(R)^+ . \tag{2.2.1}$$

This will yield an extended external product

$$K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S)$$

for all i, $j \geq 0$. Moreover, this external product should be coherent with the isomorphism

$$K_i(R) \xrightarrow{\cong} K_{i+1}(\Sigma R) \tag{2.2.2}$$

promised by the **Gersten-Wagoner delooping** [Ger72; Wag72]:

$$K_R^+ \simeq \Omega K_{\Sigma R}^+. \tag{2.2.3}$$

At this point we need the closed form of the isomorphism in Equation (2.2.2). Define the element $\tau \in GL(\Sigma\mathbb{Z}) = GL(1, \Sigma\mathbb{Z})$ by

$$\tau := \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \tag{2.2.4}$$

The isomorphism in Equation (2.2.2) is given by

$$\begin{cases} [f] \in K_i(R) & \mapsto & [f] \star \{\tau\} & \text{if } i \geq 1 \text{ ([Lod76, Corollary 2.3.6])}, \\ [P] \in K_0(R) & \mapsto & [P]\#\{\tau\} := \{p \otimes \tau + (1-p) \otimes 1\} & \text{if } i = 0 \text{ ([Lod76, page 328])}, \end{cases} \tag{2.2.5}$$

where p is the projection operator associated to the finitely generated projective R-modules P. Equally important, the extended external product should be related to the classical products

$$\# : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S) \tag{2.2.6}$$

defined by Milnor [Mil72] for i, $j \geq 0$, and $i + j \leq 2$.

There is an obvious way to extend the pairing—tensor product of modules gives a group homomorphism

$$K_0(R) \otimes K_0(S) \to K_0(R \otimes S).$$

So one might use this to extend the pairing map

$$\gamma_{\mathrm{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

by defining

$$\begin{aligned} \big(K_0(R) \times BGL(R)^+\big) \wedge \big(K_0(S) \times BGL(S)^+\big) &\to K_0(R \otimes S) \times BGL(R \otimes S)^+ \\ ([P], x) \wedge ([Q], y) &\mapsto ([P \otimes Q], \gamma_{\mathrm{Loday}}(x, y)). \end{aligned}$$

This does not work well, because the induced product map

$$K_0(R) \otimes K_i(S) \to K_i(R \otimes S)$$

is the zero map for all $i > 0$.

Here, we propose a method to extend the original pairing and examine its properties.

Definition 2.5 (The Extended Loday pairing γ'_{Loday})
The map

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{tikzcd} K_R^+ \wedge K_S^+ \arrow[r, dashed, "\gamma'_{\mathrm{Loday}}"] \arrow[d, "\simeq"'] & K_{R\otimes S}^+ \arrow[d, "\simeq"] \\ \Omega BGL(\Sigma R)^+ \wedge \Omega BGL(\Sigma S)^+ \arrow[r, "\Omega^2 \gamma_{\mathrm{Loday}}"'] & \Omega^2 BGL\big(\Sigma^2 (R \otimes S)\big)^+. \end{tikzcd} \tag{2.2.7}$$

Furthermore, we define a new product map

$$\star' : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S), \tag{2.2.8}$$

for all $i, j \geq 0$ by

$$[f] \star' [g] := \left[\gamma'_{\mathrm{Loday}} \circ (f \wedge g)\right], \tag{2.2.9}$$

Corollary 2.6 (Cf [Lod76, Proposition 2.1.8 on page 334])
Let R, S be rings, the pairing map

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+$$

defined in Definition 2.5 is

(i) natural in R and S;

(ii) bilinear;
(iii) associative;
(iv) commutative;

up to weak homotopy.

The two products are related in the following way:

Proposition 2.7
For each $i, j \geq 1$, if $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then

$$[f] \star' [g] = [f] \star \left((-1)^j [g]\right). \tag{2.2.10}$$

In particular, for $\{u\} \in K_1(R)$ and $\{v\} \in K_1(S)$, the $\star'$-product is given by the Steinberg symbol

$$\{u\} \star' \{v\} = \{u, v\}_{\mathrm{St}}. \tag{2.2.11}$$

Proof. We consider the following commutative diagram

$$\begin{tikzcd}
K_i(R) \otimes K_j(S) \arrow[r, "\star'"] \arrow[d, "(-\star\{\tau\}) \otimes (-\star\{\tau\})"', "\cong"] & K_{i+j}(R \otimes S) \arrow[d, "\cong"', "(-\star\{\tau\}) \star \{\tau\}"] \\
K_{i+1}(\Sigma R) \otimes K_{j+1}(\Sigma S) \arrow[r, "\star"'] & K_{i+j+2}\left(\Sigma^2 (R \otimes S)\right)
\end{tikzcd} \tag{2.2.12}$$

induced by Diagram (2.2.7).

If $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then the lower part of the diagram gives

$$\begin{aligned}
([f] \star \{\tau\}) \star ([g] \star \{\tau\}) &= [f] \star (\{\tau\} \star [g]) \star \{\tau\} && \text{(associativity)} \\
&= [f] \star \left((-1)^j ([g] \star \{\tau\})\right) \star \{\tau\} && \text{(graded-commutativity)} \\
&= \left([f] \star \left((-1)^j [g]\right) \star \{\tau\}\right) \star \{\tau\}. && \text{(associativity)}
\end{aligned}$$

So we must have

$$[f] \star' [g] = [f] \star \left((-1)^j [g]\right).$$

□

2.3. **Properties of the Extended Loday Pairing.**

2.3.1. *Relationship With the Classical Pairings in Algebraic K-Theory.*

Milnor defined multiplication structure on lower K-groups

$$\# : K_i(R) \otimes K_j(R) \to K_{i+j}(R \otimes R') \tag{2.3.1}$$

for i, $j \geq 0$, and $i + j \leq 2$ in [Mil72]. Actually, Milnor only defined the multiplication internally (i.e., when $R = R'$, and is commutative). But one can mimic his definition to obtain an external product as in Equation (2.3.1), so that when $R = R'$ it becomes Milnor's version.

The multiplication in Equation (2.3.1) is given by the formula:

$$\begin{cases} [P]\#[Q] &= [P \otimes Q] & \text{if } i = j = 0, \\ [P]\,\#\,\{u\} &= \{p \otimes u + (1-p) \otimes 1\} & \text{if } i = 0, \text{ and } j = 1, \\ \{u\}\,\#\,\{v\} &= \{u, v\}_{\mathrm{St}} & \text{if } i = j = 1. \end{cases} \tag{2.3.2}$$

Here, p is the idempotent operator associated to the projective module P. We omit the case $i = 0$, $j = 2$ here. The main tool we need is

Proposition 2.8 (**[Mil72, Lemma 8.9 on page 70]**)
The multiplication $\#$ in Equation (2.3.1) is associative and bilinear for i, $j > 0$, and $i + j \leq 2$.

It is also straight forward to verify that $\#$ is graded-commutative. We now relate $\#$ and $\star'$.

Theorem 2.9
For every $[P] \in K_0(R)$, $[Q] \in K_0(R')$, $\{u\} \in K_1(R)$, and $\{v\} \in K_1(R')$, we have

(i) $[P] \star' [Q] = [P]\#[Q]$,

(ii) $[P] \star' \{v\} = [P]\#(-\{v\})$,

(iii) $\{u\} \star' \{v\} = \{u\}\,\#\,\{v\}$.

Proof. We refer readers to Equation (2.2.5), and the commutative diagram in Equation (2.2.12).

(i) When $i = j = 0$, the Diagram (2.2.12) becomes

$$\begin{tikzcd} K_0(R) \otimes K_0(R') \arrow[r, "\star'"] \arrow[d, "(-\#\{\tau\}) \otimes (-\#\{\tau\})"', "\cong"] & K_0(R \otimes R') \arrow[d, "\cong"', "((-\#\{\tau\}) \star \{\tau\})"] \\ K_1(\Sigma R) \otimes K_1(\Sigma R') \arrow[r, "\star"'] & K_2(\Sigma^2(R \otimes R')). \end{tikzcd} \tag{2.3.3}$$

We then compute

$$\begin{aligned} ([P]\#\{\tau\}) \star ([Q]\#\{\tau\}) &= -\{[P]\#\{\tau\}, [Q]\#\{\tau\}\}_{\mathrm{St}} && \text{(Equation (2.1.11))} \\ &= \{[P]\#\{\tau\}, (-[Q])\,\#\,\{\tau\}\}_{\mathrm{St}} && \text{(Bilinearity of } \{-,-\}_{\mathrm{St}}) \\ &= ([P]\#\{\tau\})\,\#\,((-[Q])\,\#\,\{\tau\}) && \text{(Equation (2.3.2))} \\ &= [P]\#(\{\tau\}\,\#\,(-[Q]))\,\#\,\{\tau\} && \text{(Associativity of } \#) \end{aligned}$$

$$
\begin{aligned}
&= [P]\#\left(-[Q]\#\{\tau\}\right)\#\{\tau\} && \text{(Graded-commutativity of } \#) \\
&= -\left([P]\#\left([Q]\#\{\tau\}\right)\#\{\tau\}\right) && \text{(Bilinearity of } \#) \\
&= -\left\{[P]\#\left([Q]\#\{\tau\}\right),\tau\right\}_{\mathrm{St}} && \text{(Equation (2.3.2))} \\
&= \left([P]\#\left([Q]\#\{\tau\}\right)\right)\star\{\tau\} && \text{(Equation (2.1.11))} \\
&= \left(\left([P]\#[Q]\right)\#\{\tau\}\right)\star\{\tau\}, && \text{(Associativity of } \#)
\end{aligned}
$$

proving

$$[P]\star'[Q] = [P]\#[Q].$$

(ii) When $i = 0$, $j = 1$, Diagram (2.2.12) becomes

$$
\begin{tikzcd}
K_0(R)\otimes K_1(R') \arrow[r, "\star'"] \arrow[d, "(-\#\{\tau\})\otimes(-\star\{\tau\})"', "\cong"] & K_1(R\otimes R') \arrow[d, "\cong"', "((-\star\{\tau\})\star\{\tau\})"] \\
K_1(\Sigma R)\otimes K_2(\Sigma R') \arrow[r, "\star"'] & K_3(\Sigma^2(R\otimes R')).
\end{tikzcd}
\tag{2.3.4}
$$

We then compute

$$
\begin{aligned}
\left([P]\#\{\tau\}\right)\star\left(\{v\}\star\{\tau\}\right) &= \left(\left([P]\#\{\tau\}\right)\star\{v\}\right)\star\{\tau\} && \text{(Associativity of } \star) \\
&= \left(-\left\{[P]\#\{\tau\},v\right\}_{\mathrm{St}}\right)\star\{\tau\} && \text{(Proposition 2.4)} \\
&= \left(-\left(\left([P]\#\{\tau\}\right)\#\{v\}\right)\right)\star\{\tau\} && \text{(Equation (2.3.2))} \\
&= \left(-\left([P]\#\left(\{\tau\}\#\{v\}\right)\right)\right)\star\{\tau\} && \text{(Associativity of } \#) \\
&= \left([P]\#\left(\{v\}\#\{\tau\}\right)\right)\star\tau && \text{(Graded-commutativity of } \#) \\
&= \left(\left([P]\#\{v\}\right)\#\{\tau\}\right)\star\{\tau\} && \text{(Associativity of } \#) \\
&= \left\{[P]\#\{v\},\{\tau\}\right\}_{\mathrm{St}}\star\{\tau\} && \text{(Equation (2.3.2))} \\
&= \left(-\left(\left([P]\#\{v\}\right)\star\{\tau\}\right)\right)\star\{\tau\} && \text{(Proposition 2.4)}
\end{aligned}
$$

$$= ((-([P]\#\{v\}) \star \{\tau\})) \star \{\tau\} \qquad \text{(Bilinearity of } \star\text{)}$$

$$= (([P]\#(-\{v\})) \star \{\tau\}) \star \{\tau\} \qquad \text{(Bilinearity of } \#\text{)},$$

proving

$$[P] \star' \{v\} = [P]\#(-\{v\}).$$

(iii)

$$\{u\} \star' \{v\} = \{u, v\}_{\mathrm{St}} \qquad \text{(Proposition 2.7)}$$

$$= \{u\} \# \{v\}.$$

□

The following case is still unknown.

Question 2.10
Is the product map

$$\star' : K_0(R) \otimes K_2(S) \to K_2(R \otimes S) \tag{2.3.5}$$

compatible with classical product map

$$\# : K_0(R) \otimes K_2(S) \to K_2(R \otimes S). \tag{2.3.6}$$

defined by Milnor in [Mil72, page 51]?

2.3.2. *The Non-Connective Gersten-Wagoner Algebraic K-Theory Spectrum* $\mathbb{K}_R^{\mathrm{GW}}$.

We relate our extended Loday pairing

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+$$

to the structure map

$$\mathbb{S}^1 \wedge K_R^+ \to K_{\Sigma R}^+$$

of the Gersten-Wagoner spectrum $\mathbb{K}_R^{\mathrm{GW}}$. This will be used later in proving our version of the Loday assembly is a map of spectra.

In [Lod76, page 341–343], Loday gave an explicit description of the Gersten-Wagoner delooping

$$K_R^+ \xrightarrow{\simeq} \Omega K_{\Sigma R}^+,$$

so that the induced isomorphisms on homotopy groups in Equation (2.2.2) are given as in Equation (2.2.5). We will use our extended Loday pairing to define a map

$$\mathbb{S}^1 \wedge K_R^+ \to K_{\Sigma R}^+,$$

whose adjoint

$$K_R^+ \to \Omega K_{\Sigma R}^+$$

will induce isomorphisms on homotopy groups, and hence is a homotopy equivalence by the Whitehead Theorem. We begin by thinking the circle $\mathbb{S}^1$ as the classifying space $B\langle t\rangle$ of the infinite cyclic group generated by t, and define the following group homomorphism:

Definition 2.11 (The map $\mathbf{t}^+ : B\langle t\rangle \to K_{\Sigma\mathbb{Z}}^+$)
Let $\langle t\rangle$ be the infinite cyclic group generated by t, and $\tau \in GL(\Sigma\mathbb{Z})$ be the element defined in Equation (2.2.4). Define the group homomorphism

$$\begin{aligned} \langle t\rangle &\to GL(\Sigma\mathbb{Z}) \\ t &\mapsto \tau^{-1}. \end{aligned}$$

The induced map

$$B\langle t\rangle \to K_{\Sigma\mathbb{Z}}^+ \tag{2.3.7}$$

is denoted by $\mathbf{t}^+$.

Proposition 2.12 (Another Gersten-Wagoner Delooping)
Let R be a ring, and $\langle t\rangle$ be the infinite cyclic group generated by t. The adjoint of the composition

$$B\langle t\rangle \wedge K_R^+ \xrightarrow{\mathbf{t}^+\wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma R}^+ \tag{2.3.8}$$

is a weak equivalence, and hence a homotopy equivalence.

Proof. A class $[f] \in K_i(R)$ is represented by a spheroid

$$f : \mathbb{S}^i \to K_R^+.$$

Suspending it yields the following composition

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\mathrm{id}\wedge f} \mathbb{S}^1 \wedge K_R^+ \xrightarrow{\mathbf{t}^+\wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma R}^+, \tag{2.3.9}$$

which represents the element $[f] \star' \left\{\tau^{-1}\right\} \in K_{i+1}(\Sigma R)$. In particular, we have

$$[f] \star' \left\{\tau^{-1}\right\} = \begin{cases} [f] \star \{\tau\} & \text{if } i > 0 \quad (\text{Proposition 2.7}) \\ [f] \,\#\, \{\tau\} & \text{if } i = 0 \quad (\text{Theorem 2.9 (ii)}) \end{cases} \tag{2.3.10}$$

The map on homotopy groups

$$K_i(R) \to K_{i+1}(\Sigma R)$$

induced by the adjoint of (2.3.8) sends

$$[f] \mapsto [f] \star' \left\{\tau^{-1}\right\},$$

which is an isomorphism for $i = 0$ (resp. $i > 0$) by [Lod76, after Theorem 1.4.7 on page 328] (resp. [Lod76, Corollary 2.3.6 on page 345]). Therefore, Whitehead Theorem asserts the homotopy equivalence

$$K_R^+ \simeq K_{\Sigma R}^+.$$

□

Consequently, we have the following Ω-spectrum

Definition 2.13 (The Non-connective Gersten-Wagoner Algebraic K-theory Spectrum $\mathbb{K}_R^{\mathrm{GW}}$)
Let R be a ring. Define the spectrum $\mathbb{K}_R^{\mathrm{GW}}$ by having n-th space as

$$\left(\mathbb{K}_R^{\mathrm{GW}}\right)_n := \begin{cases} K_{\Sigma^n R}^+ & \text{if } n \geq 0, \\ \Omega^{-n} K_R^+ & \text{if } n < 0, \end{cases} \tag{2.3.11}$$

with the convention that $\Sigma^0 R := R$. The structure maps

$$f_n : \left(\mathbb{K}_R^{\mathrm{GW}}\right)_n \to \left(\mathbb{K}_R^{\mathrm{GW}}\right)_{n+1} \tag{2.3.12}$$

are given by

$$f_n := \begin{cases} (2.3.8) & \text{if } n \geq 0, \\ \text{adjoint of } \mathrm{id}_{\left(\mathbb{K}_R^{\mathrm{GW}}\right)_n} & \text{if } n < 0. \end{cases} \tag{2.3.13}$$

By construction, $\mathbb{K}_R^{\mathrm{GW}}$ is an Ω-spectrum.

3. The Main Theorem

3.1. **Definition of the Loday Assembly.** We use the extended Loday pairing γ'_{Loday} to define the Loday assembly map:

Definition 3.1 (The Loday Assembly Map α_{Loday} in Reduced Case)
Let R be a ring, and G be a group. The **Loday assembly map** is the map of spectra

$$\alpha_{\mathrm{Loday}} : BG \wedge \mathbb{K}_R^{\mathrm{GW}} \to \mathbb{K}_{R[G]}^{\mathrm{GW}}, \tag{3.1.1}$$

whose components are given by the composition of maps of spaces:

$$BG \wedge K_{\Sigma^m R}^+ \xrightarrow{j^+ \wedge \mathrm{id}} K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma^m R}^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma^m R[G]}^+. \tag{3.1.2}$$

Here, the map

$$j^+ : BG \to K_{\mathbb{Z}[G]}^+ \tag{3.1.3}$$

is induced by the group homomorphism

$$\begin{aligned} j : G &\to GL\left(\mathbb{Z}[G]\right) \\ g &\mapsto \begin{bmatrix} g & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \end{aligned} \tag{3.1.4}$$

We need to justify our definition.

Proposition 3.2 (Cf [Lod76, Proposition 4.1.1 on page 356])
Let R be a ring, and G be a group. The Loday assembly map

$$\alpha_{\text{Loday}} : BG \wedge \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}}$$

defined in Definition 3.1 is a map of spectra, well-defined up to weak homotopy.

Proof. We need to show the components of α_{Loday}, as defined in Equation (3.1.2), commute with the structure maps of $\mathbb{K}_R^{\text{GW}}$ as defined in Equation (2.3.8). Let $\langle t \rangle$ be the infinite cyclic group generated by t. By unwrapping the definitions of α_{Loday} and the structure maps given in Proposition 2.12, the following diagram

$$\begin{tikzcd}
BG \wedge B\langle t \rangle \wedge K_R^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \mathrm{t}^+ \wedge \mathrm{id}"'] & K_{\mathbb{Z}[G]}^+ \wedge B\langle t \rangle \wedge K_R^+ \arrow[r, "\text{twist}", "\cong"'] \arrow[d, "\mathrm{id} \wedge \mathrm{t}^+ \wedge \mathrm{id}"'] & B\langle t \rangle \wedge K_{\mathbb{Z}[G]}^+ \wedge K_R^+ \arrow[r, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"] \arrow[d, "\mathrm{t}^+ \wedge \mathrm{id} \wedge \mathrm{id}"'] & B\langle t \rangle \wedge K_{R[G]}^+ \arrow[d, "\mathrm{t}^+ \wedge \mathrm{id}"] \\
BG \wedge K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"'] & K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \arrow[r, "\cong", "\text{twist}"'] \arrow[d, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"'] & K_{\Sigma\mathbb{Z}}^+ \wedge K_{\mathbb{Z}[G]}^+ \wedge K_R^+ \arrow[r, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"] \arrow[d, "\gamma'_{\text{Loday}} \circ \gamma'_{\text{Loday}}"'] & K_{\Sigma\mathbb{Z}}^+ \wedge K_{R[G]}^+ \arrow[dl, "\gamma'_{\text{Loday}}"] \\
BG \wedge K_{\Sigma R}^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"'] & K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma R}^+ \arrow[r, "\gamma'_{\text{Loday}}"'] & K_{\Sigma R[G]}^+ &
\end{tikzcd} \tag{3.1.5}$$

commutes up to weak homotopy by Corollary 2.6. Here, the columns are the structure maps, and the rows are components of the Loday assembly. □

The homotopy groups $\pi_i(BG \wedge \mathbb{K}_R^{\text{GW}})$ of the spectrum $BG \wedge \mathbb{K}_R^{\text{GW}}$ is the reduced homology groups of the classifying space BG of G with coefficients in $\mathbb{K}_R^{\text{GW}}$. So the induced map

$$\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG \wedge \mathbb{K}_R^{\text{GW}}) \to K_i(R[G])$$

approximates the algebraic K-theory of the group ring $R[G]$ by a reduced homology theory. We want to lift the domain to an unreduced homology theory. Recall for any (based) topological space X, there is a cofibre sequence

$$X \to X_+ \to \mathbb{S}^0 \tag{3.1.6}$$

of spaces, where $X_+ := X \sqcup \text{point}$ is the space obtained by adding a disjoint point to X. If $\mathbb{E}$ is a spectrum, then we have a cofibre sequence

$$X \wedge \mathbb{E} \to X_+ \wedge \mathbb{E} \to \mathbb{S}^0 \wedge \mathbb{E} \tag{3.1.7}$$

of spectra. Consequently, we have a splitting

$$\begin{aligned} X_+ \wedge \mathbb{E} &\simeq (X \wedge \mathbb{E}) \vee \left(\mathbb{S}^0 \wedge \mathbb{E}\right) \\ &\cong (X \wedge \mathbb{E}) \vee \mathbb{E} \end{aligned} \tag{3.1.8}$$

of spectra. This allows us to extend the Loday assembly to the unreduced case.

Definition 3.3 (The Loday Assembly Map α_{Loday} in Unreduced Case) Let R be a ring and G be a group. The canonical inclusion $R \to R[G]$ induces a map

$$i : \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}} \tag{3.1.9}$$

of spectra. The Loday assembly in unreduced case is defined by

$$BG_+ \wedge \mathbb{K}_R^{\text{GW}} \simeq \left(BG \wedge \mathbb{K}_R^{\text{GW}}\right) \vee \mathbb{K}_R^{\text{GW}} \xrightarrow{\text{(Definiton 3.1)}\vee i} \mathbb{K}_{R[G]}^{\text{GW}}. \tag{3.1.10}$$

By abuse of terminology and notation, we call this map the Loday assembly and denote it by α_{Loday} as well

3.2. **A Subgroup of the Source of Loday Assembly.** The source of the Loday assembly

$$\pi_i(\alpha_{\text{Loday}}) : \pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right) \to K_{i+1}\left(R[G]\right) \tag{3.2.1}$$

on homotopy groups can be computed by the **Atiyah-Hirzebruch spectral sequence**:

$$\begin{aligned}\text{AHSS}\left(BG\right)_{p,q}^2 &\cong H_p\left(BG; K_q(R)\right)\\ &\Rightarrow \pi_{p+q}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right),\end{aligned} \tag{3.2.2}$$

and in particular,

$$\begin{aligned}E_{1,i}^2 &\cong H_1\left(BG; K_i(R)\right)\\ &\cong G_{ab} \otimes K_i(R).\end{aligned} \tag{3.2.3}$$

Because the Atiyah-Hirzebruch spectral sequence is concentrated on the right-half plane, it follows that the differential

$$d_{1,i}^2 : E_{1,i}^2 \to E_{-1,i+1}^2 \tag{3.2.4}$$

has to be zero. So the group $E_{1,i}^\infty$ is a quotient of $E_{1,i}^2$, and we shall see it is also a subgroup of the source $\pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right)$.

Fix a spectrum $\mathbb{E}$. For any CW complex X, we have a homeomorphism

$$X_+ \cong X \vee \mathbb{S}^0, \tag{3.2.5}$$

and hence an isomorphism of homotopy groups

$$\pi_{i+1}(X_+ \wedge \mathbb{E}) \cong \pi_{i+1}(X \wedge \mathbb{E}) \oplus \pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E}). \tag{3.2.6}$$

Fix a skeleta filtration

$$\text{point} = X^{(0)} \subseteq X^{(1)} \subseteq \cdots \subseteq X \tag{3.2.7}$$

on X, the identification in Equation (3.2.5) gives a skeleta filtration

$$X_+^{(p)} \cong X^{(p)} \vee \mathbb{S}^0 \tag{3.2.8}$$

on X_+. Thus we have a splitting for the Atiyah-Hirzebruch spectral sequence:

$$\text{AHSS}\left(X_+\right)_{p,q}^r \cong \text{AHSS}\left(X\right)_{p,q}^r \oplus \text{AHSS}\left(\mathbb{S}^0\right)_{p,q}^r \tag{3.2.9}$$

for $r \geq 1$. Tracing through the construction of the Atiyah-Hirzebruch spectral sequence (see, for example [Wic15]), we see that the splitting in Equation (3.2.5) respects the filtration on homotopy groups:

$$\mathcal{F}_1 \left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] \cong \mathcal{F}_1 \left[\pi_{i+1}(X \wedge \mathbb{E})\right] \oplus \mathcal{F}_1 \left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})\right], \tag{3.2.10}$$

for which

$$\mathcal{F}_1 \left[\pi_{i+1}\left(X_+ \wedge \mathbb{E}\right)\right] := \operatorname{im}\left(\pi_{i+1}\left(X_+^{(1)} \wedge \mathbb{E}\right) \to \pi_{i+1}\left(X_+ \wedge \mathbb{E}\right)\right). \tag{3.2.11}$$

The subgroup

$$\mathcal{F}_1 \left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] \subseteq \pi_{i+1}(X_+ \wedge \mathbb{E}) \tag{3.2.12}$$

is what we are interested in, and it can be identified as:

$$\begin{aligned} \mathcal{F}_1 \left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] &\cong \mathcal{F}_1 \left[\pi_{i+1}(X \wedge \mathbb{E})\right] \oplus \mathcal{F}_1 \left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})\right] \\ &= \mathrm{AHSS}\,(X)^{\infty}_{1,i} \oplus \pi_{i+1}(\mathbb{E}). \end{aligned} \tag{3.2.13}$$

3.3. **Statement of the Main Theorem and the Proof.** Now, let us work with $\mathbb{E} = \mathbb{K}_R^{\mathrm{GW}}$, and $X = BG$. Then Equation (3.2.13) says

$$\begin{aligned} \mathcal{F}_1 \left[\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}})\right] &\cong \mathcal{F}_1 \left[\pi_{i+1}(BG \wedge \mathbb{K}_R^{\mathrm{GW}})\right] \oplus \mathcal{F}_1 \left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{K}_R^{\mathrm{GW}})\right] \\ &= \mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus \pi_{i+1}(\mathbb{K}_R^{\mathrm{GW}}) \\ &= \mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus K_{i+1}(R). \end{aligned} \tag{3.3.1}$$

So the source of the Loday assembly

$$\alpha_{\mathrm{Loday}} : \pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \to K_{i+1}(R[G])$$

contains

$$\mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus K_{i+1}(R)$$

as a subgroup. In fact, we already know what the Loday assembly is doing when restricted onto the summand $K_{i+1}(R)$—it is induced by the inclusion

$$i : R \to R[G]$$

of rings. See [Lod76, before Lemma 4.1.2 on page 356], or Definition 3.3 before. What we want now is to study the Loday assembly when restricted onto the summand $\mathrm{AHSS}\,(BG)^{\infty}_{1,i}$.

The splitting of the Atiyah-Hirzebruch spectral sequence in Equation (3.2.9) says the differentials of $\mathrm{AHSS}\,(BG_+)^2_{p,q}$ splits into two parts, which we illustrate here:

$$
\underset{\mathrm{AHSS}(BG_+)^2_{p,q}}{\begin{array}{c|cccc}
q & & & & \\
& K_5(R) & H_1(BG;K_5(R)) & H_2(BG;K_5(R)) & \cdots \\
& K_4(R) & H_1(BG;K_4(R)) & H_2(BG;K_4(R)) & \cdots \\
& K_3(R) & H_1(BG;K_3(R)) & H_2(BG;K_3(R)) & \cdots \\
& K_2(R) & H_1(BG;K_2(R)) & H_2(BG;K_2(R)) & \cdots \\
& K_1(R) & H_1(BG;K_1(R)) & H_2(BG;K_1(R)) & \cdots \\
& K_0(R) & H_1(BG;K_0(R)) & H_2(BG;K_0(R)) & \cdots \\
\hline
& & & & p
\end{array}}
\cong
\underset{\mathrm{AHSS}(BG)^2_{p,q}}{\begin{array}{c|cccc}
q & & & & \\
& 0 & H_1(BG;K_5(R)) & H_2(BG;K_5(R)) & \cdots \\
& 0 & H_1(BG;K_4(R)) & H_2(BG;K_4(R)) & \cdots \\
& 0 & H_1(BG;K_3(R)) & H_2(BG;K_3(R)) & \cdots \\
& 0 & H_1(BG;K_2(R)) & H_2(BG;K_2(R)) & \cdots \\
& 0 & H_1(BG;K_1(R)) & H_2(BG;K_1(R)) & \cdots \\
& 0 & H_1(BG;K_0(R)) & H_2(BG;K_0(R)) & \cdots \\
\hline
& & & & p
\end{array}}
\oplus
\underset{\mathrm{AHSS}(\mathbb{S}^0)^2_{p,q}}{\begin{array}{c|cccc}
q & & & & \\
& K_5(R) & 0 & 0 & \cdots \\
& K_4(R) & 0 & 0 & \cdots \\
& K_3(R) & 0 & 0 & \cdots \\
& K_2(R) & 0 & 0 & \cdots \\
& K_1(R) & 0 & 0 & \cdots \\
& K_0(R) & 0 & 0 & \cdots \\
\hline
& & & & p
\end{array}}
\tag{3.3.2}
$$

(We omit the negative K-groups because they are annoying to include.) We note that the groups $\mathrm{AHSS}\,(BG)^2_{0,q}$ are trivial because the reduced homology group $\widetilde{H}_0(BG)$ is trivial for BG being path-connected. Consequently,

Lemma 3.4
Let G be a group, and R be a ring. The differentials

$$d^2_{1,q} : \mathrm{AHSS}\,(BG_+)^2_{1,q} \to \mathrm{AHSS}\,(BG_+)^2_{-1,q+1} \tag{3.3.3}$$

in the E^2-page of the Atiyah-Hirzebruch spectral sequence of BG_+ associated to the K-theory spectrum $\mathbb{K}^{\mathrm{GW}}_R$ are trivial for all q.

Proof. The splitting of the Atiyah-Hirzebruch spectral sequence in Equation (3.2.9) says the differential

$$d^2_{1,q} : \mathrm{AHSS}\,(BG_+)^2_{1,q} \to \mathrm{AHSS}\,(BG_+)^2_{-1,q+1}$$

splits into direct sum of the two differentials

$$\mathrm{AHSS}\,(BG)^2_{1,q} \to \mathrm{AHSS}\,(BG)^2_{-1,q+1}\,, \tag{3.3.4}$$

$$\mathrm{AHSS}\,(\mathbb{S}^0)^2_{1,q} \to \mathrm{AHSS}\,(\mathbb{S}^0)^2_{-1,q+1}\,. \tag{3.3.5}$$

From Diagram (3.3.2), we see that the first (resp. second) map is trivial because the codomain (resp. domain) is trivial, hence proving the claim. □

Lemma 3.4 says the group $\mathrm{AHSS}\,(BG_+)^\infty_{1,i}$ is a quotient of $\mathrm{AHSS}\,(BG_+)^2_{1,i}$, and therefore we have a diagram

$$\begin{tikzcd} \mathrm{AHSS}\,(BG)^2_{1,i} \cong G_{ab} \otimes K_i(R) \arrow[r, dashed] \arrow[d, two heads] & K_{i+1}\left(R[G]\right) \\ \mathrm{AHSS}\,(BG)^\infty_{1,i} \arrow[r, hook] & \pi_{i+1}\left(BG \wedge \mathbb{K}^{\mathrm{GW}}_R\right). \arrow[u, "\alpha_{\mathrm{Loday}}"'] \end{tikzcd} \tag{3.3.6}$$

We would like to study the filler. (Perhaps we should explain the absent of the disjoint base-point "+" in Diagram (3.3.6) is not a typo. As pointed out in Equation (3.3.1), $\pi_{i+1}(BG_+ \wedge \mathbb{K}^{\mathrm{GW}}_R)$ contains $K_{i+1}(R) \oplus \mathrm{AHSS}\,(BG)^\infty_{1,i}$ as a subgroup, and we want to know what the Loday assembly is doing on the term $\mathrm{AHSS}\,(BG)^\infty_{1,i}$.)

Let us go back to the E^1-page of the Atiyah-Hirzebruch spectral sequence. Unlike the later pages, the E^1-page is not homotopy invariant—it depends on the choice of cellular structure on BG. **For our purpose, we are using the skeletal filtration on the bar construction for $\boldsymbol{BG}$**, so that

$$\mathrm{AHSS}\,(BG)^1_{1,i} \cong \pi_{i+1}\left(\left(\bigvee_{g\in G} \mathbb{S}^1\right) \wedge \mathbb{K}^{\mathrm{GW}}_R\right). \tag{3.3.7}$$

The filler we want to study is then induced by the composition

$$\pi_{i+1}\left(\left(\bigvee_{g\in G} \mathbb{S}^1\right) \wedge \mathbb{K}^{\mathrm{GW}}_R\right) \xrightarrow{(\mathfrak{i}_G)_*} \pi_{i+1}\left(BG \wedge \mathbb{K}^{\mathrm{GW}}_R\right) \xrightarrow{\pi_{i+1}\left(\alpha_{\mathrm{Loday}}\right)} K_{i+1}\left(R[G]\right), \tag{3.3.8}$$

for which the first arrow is induced by the inclusion

$$\mathfrak{i}_G : \bigvee_{g\in G} \mathbb{S}^1 \hookrightarrow BG \tag{3.3.9}$$

of the 1-skeleton into BG. This follows from the construction of the Atiyah-Hirzebruch spectral sequence. (See [Wic15] for example.) When $G = \langle t \rangle$ is the infinite cyclic group, we know the circle $\mathbb{S}^1$ is a model for $B\langle t \rangle$. Moreover,

Lemma 3.5
The inclusion

$$\mathfrak{i}_t : \mathbb{S}^1 \to B\langle t \rangle \tag{3.3.10}$$

that sends $\mathbb{S}^1$ to the 1-cell of $B\langle t \rangle$ labelled by the generator t is a weak equivalence.

Proof. It is clear that the induced group homomorphism

$$(\mathfrak{i}_t)_* : \pi_i\left(\mathbb{S}^1\right) \to \pi_i\left(B\langle t \rangle\right) \tag{3.3.11}$$

is an isomorphism for $i \neq 1$, because the source and target are both trivial. So we need to check the map

$$(\mathfrak{i}_t)_* : \pi_1\left(\mathbb{S}^1\right) \to \pi_1\left(B\langle t \rangle\right) \tag{3.3.12}$$

is an isomorphism.

The map in Equation (3.3.12) is just an endomorphism of infinite cyclic group. In particular, it is surjective by the definition of $\mathfrak{i}_t$. (The generator is in the image.) So the First Isomorphism Theorem says the map in Equation (3.3.12) must have trivial kernel, and hence an isomorphism. □

The Whitehead Theorem then asserts $\mathfrak{i}_t$ is a homotopy equivalence. Thus when $G = \langle t \rangle$ is the infinite cyclic group, we have the following commutative diagram

$$\begin{tikzcd}
\bigvee_{g \in \langle t \rangle} \mathbb{S}^1 \arrow[r, "\mathfrak{i}_{\langle t \rangle}"] & B\langle t \rangle \\
& \mathbb{S}^1 \arrow[ul, "\mathfrak{i}_t"] \arrow[u, "\mathfrak{i}_t"', "\simeq"]
\end{tikzcd} \tag{3.3.13}$$

The upshot is that we are able to describe the composition in Equation (3.3.8) when $G = \langle t \rangle$.

Proposition 3.6
When $G = \langle t \rangle$ is the infinite cyclic group, the composition in Equation (3.3.8) can be identified as

$$\begin{tikzcd}
\pi_{i+1}\left(\left(\bigvee_{g \in \langle t \rangle} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\mathfrak{i}_{\langle t \rangle})"] & \pi_{i+1}\left(B\langle t \rangle \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "\pi_{i+1}(\alpha_{\mathrm{Loday}})"] & K_{i+1}\left(R[t^{\pm 1}]\right) \\
\pi_{i+1}\left(\mathbb{S}^1 \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[u, "(\mathfrak{i}_t)_*"] \arrow[r, "\mathrm{id}"'] & \pi_{i+1}\left(\mathbb{S}^1 \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[u, "(\mathfrak{i}_t)_*", "\cong"'] \arrow[r] & K_{i+1}\left(R[t^{\pm 1}]\right) \arrow[u, equal] \\
K_i(R) \arrow[u, "\cong"] \arrow[r, "\mathrm{id}"'] & K_i(R) \arrow[u, "\cong"] \arrow[r] & K_{i+1}\left(R[t^{\pm 1}]\right), \arrow[u, equal]
\end{tikzcd} \tag{3.3.14}$$

where the bottom composition sends $[f] \in K_i(R)$ to

$$[f] \star \{\mu_{t^{-1}}\} \in K_{i+1}\left(R[t^{\pm 1}]\right),$$

for which $\mu_{t^{-1}} \in \mathrm{Aut}\left(R[t^{\pm 1}]\right)$ is multiplication by t^{-1}.

We remind readers the commutativity of the top-left square comes from Diagram (3.3.13).

Proof. The class $[f] \in K_i(R)$ is represented by the spheroid

$$f : \mathbb{S}^i \to K_R^+,$$

where

$$K_R^+ := K_0(R) \times BGL(R)^+.$$

Taking suspension yields

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\mathrm{id} \wedge f} \mathbb{S}^1 \wedge K_R^+ \xrightarrow{j^+ \wedge \mathrm{id}} K_{\mathbb{Z}[t^{\pm 1}]}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\text{Loday}}} K_{R[t^{\pm 1}]}^+.$$

This composition represents the element $\pi_{i+1}(\alpha_{\text{Loday}})([f]) \in K_{i+1}\left(R[t^{\pm 1}]\right)$, as well as the product

$$[f] \star' \left\{j^+\right\} \in K_{i+1}\left(R[t^{\pm 1}]\right).$$

Now, j^+ is induced by the group homomorphism

$$\begin{aligned} j : \langle t \rangle &\to GL\left(\mathbb{Z}[t^{\pm 1}]\right) \\ t &\mapsto \begin{bmatrix} t & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \end{aligned}$$

So $\left\{j^+\right\} = \{\mu_t\}$, and thus

$$\begin{aligned} [f] \star' \left\{j^+\right\} &= [f] \star' \{\mu_t\} \\ &= [f] \star (-\{\mu_t\}) && (\text{Proposition 2.7}) \\ &= [f] \star \{\mu_{t^{-1}}\} \end{aligned}$$

as desired. □

For arbitrary group G, we recall the isomorphism

$$\begin{aligned} G &\cong \operatorname{Hom}(\langle t \rangle, G) \\ g &\mapsto (\varphi_g : t \mapsto g), \end{aligned} \tag{3.3.15}$$

allowing us to extend Proposition 3.6 to arbitrary G.

Proposition 3.7

For an arbitrary group G, under the canonical isomorphism

$$\pi_{i+1}\left(\left(\bigvee_{g \in G} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\text{GW}}\right) \cong \bigoplus_{g \in G} K_i(R), \tag{3.3.16}$$

the composition in (3.3.8) can be identified as

$$\begin{tikzcd}
\pi_{i+1}\left(\left(\bigvee_{g\in G}\mathbb{S}^1\right)\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\mathfrak{i}_G)_*"] \arrow[d, phantom, "\cong"] & \pi_{i+1}\left(BG\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "\pi_{i+1}(\alpha_{\mathrm{Loday}})"] \arrow[d, equal] & K_{i+1}\left(R[G]\right) \arrow[d, equal] \\
\bigoplus_{g\in G} K_i(R) \arrow[r] & \pi_{i+1}\left(BG\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r] & K_{i+1}\left(R[G]\right),
\end{tikzcd} \tag{3.3.17}$$

where the bottom composition sends the element $[f] \in K_i(R)$ in the summand in $\bigoplus_{g\in G} K_i(R)$ labelled by $g \in G$ to the element

$$[f] \star \{\mu_{g^{-1}}\} \in K_{i+1}\left(R[G]\right),$$

for which $\mu_{g^{-1}} \in \mathrm{Aut}\left(R[G]\right)$ is multiplication by g^{-1}.

Proof. Fix $g \in G$, and consider the commutative diagram

$$\begin{tikzcd}
\pi_{i+1}\left(\mathbb{S}^1\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\varphi_g)_*"] \arrow[d, "(\mathfrak{i}_t)_*"', "\cong"] & \pi_{i+1}\left(\left(\bigvee_{g\in G}\mathbb{S}^1\right)\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[d, "(\mathfrak{i}_G)_*"] \\
\pi_{i+1}\left(B\langle t\rangle\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\varphi_g)_*"'] \arrow[d, "\pi_{i+1}(\alpha_{\mathrm{Loday}})"'] & \pi_{i+1}\left(BG\wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[d, "\pi_{i+1}(\alpha_{\mathrm{Loday}})"] \\
K_{i+1}\left(R[t^{\pm1}]\right) \arrow[r, "(\varphi_g)_*"'] & K_{i+1}\left(R[G]\right).
\end{tikzcd} \tag{3.3.18}$$

The top square commutes by chasing the definitions of the arrows. The bottom square commutes by the naturality of α_{Loday} via the group homomorphism

$$\varphi_g : \langle t\rangle \to G.$$

Now, Proposition 3.6 says the left vertical composition in Diagram (3.3.18) sends $[f] \in K_i(R)$ to

$$[f] \star \{\mu_{t^{-1}}\} \in K_{i+1}\left(R[t^{\pm1}]\right).$$

So the commutativity of Diagram (3.3.18) says the right vertical composition sends $[f] \in K_i(R)$ in the summand $\bigoplus_{g\in G} K_i(R)$ indexed by $g \in G$ to the element

$$[f] \star \{\mu_{g^{-1}}\} \in K_{i+1}\left(R[G]\right).$$

∎

Theorem 3.8 (An Explicit Formula for the Loday Assembly)
The filler in Diagram (3.3.6) is induced by the map

$$\begin{aligned} G \times K_i(R) &\to K_{i+1}(R[G]) \\ (g, [f]) &\mapsto [f] \star \{\mu_{g^{-1}}\}. \end{aligned}$$

Proof. Follows immediately from Proposition 3.7. ☐

Corollary 3.9 (The Loday Assembly on π_1, [Wal78b, Assertion 15.8 on page 229], [LR05, page 708])
For a regular ring R, and a group G, the Loday assembly on π_1 is given by

$$K_1(i) \oplus \Phi_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \to K_1(R[G]), \tag{3.3.19}$$

for which $i : R \to R[G]$ is the inclusion, and Φ_1 is induced by the map

$$\begin{aligned} G \times K_0(R) &\to K_1(R[G]) \\ (g, [P]) &\mapsto \{\mathfrak{h}_g\}, \end{aligned} \tag{3.3.20}$$

for which $\mathfrak{h}_g : P \otimes_R R[G] \to R \otimes_R R[G]$ is the automorphism given by

$$\mathfrak{h}_g(x \otimes u) := x \otimes ug^{-1}.$$

Proof. If the ring R is regular, then the Atiyah-Hirzebruch spectral sequence for $\pi_i(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}})$ is first-quadrant. Hence,

$$\pi_1(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \cong K_1(R) \oplus [G_{ab} \otimes K_0(R)].$$

If $g \in G$, and $[P] \in K_0(R)$, then Theorem 3.8 says the Loday assembly sends $(g, [P])$ to

$$\begin{aligned} [P] \star' \{\mu_g\} &= [P] \# \{\mu_{g^{-1}}\} && \text{(Theorem 2.9)} \\ &= \{\mathfrak{h}_g\}. \end{aligned}$$

☐

Corollary 3.10 ([Wal78b, Proposition 15.7 (1) on page 229])
Let G be a group. The cokernel of the Loday assembly

$$\operatorname{coker}\left(\pi_1(\alpha_{\mathrm{Loday}}) : \pi_1\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \to K_1(\mathbb{Z}[G])\right) \tag{3.3.21}$$

on π_1 is isomorphic to the first Whitehead group

$$\mathrm{Wh}_1(G) := \frac{K_1(\mathbb{Z}[G])}{\pm G}. \tag{3.3.22}$$

Proof. We need to check the image $\operatorname{im}(\pi_1(\alpha_{\mathrm{Loday}}))$ of the Loday assembly on π_1 is the subgroup $\pm G_{ab}$ of $K_1(\mathbb{Z}[G])$.

Note that the ring $\mathbb{Z}$ has no negative K-groups for being regular. Thus, the vanishing of the differentials of the Atiyah-Hirzebruch spectral sequences says

$$\begin{aligned}\pi_1\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) &\cong K_1(\mathbb{Z}) \oplus [G_{ab} \otimes K_0(\mathbb{Z})] && (3.3.23)\\ &\cong C_2 \oplus [G_{ab} \otimes \mathbb{Z}] \\ &\cong \{1,-1\} \oplus G_{ab} && (3.3.24)\end{aligned}$$

We already know, from Definition 3.3, the Loday assembly $\pi_1\left(\alpha_{\mathrm{Loday}}\right)|_{K_1(\mathbb{Z})}$ when restricted onto the summand $K_1(\mathbb{Z})$ of $\pi_1\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right)$ is the group homomorphism

$$K_1(\mathbb{Z}) \to K_1(\mathbb{Z}[G])$$

induced by the canonical inclusion $i : \mathbb{Z} \to \mathbb{Z}[G]$. Therefore, under the identification

$$K_1(\mathbb{Z}) \cong \{1,-1\},$$

the restriction $\pi_1\left(\alpha_{\mathrm{Loday}}\right)|_{K_1(\mathbb{Z})}$ sends ± 1 to $\pm 1 \in K_1(\mathbb{Z}[G])$.

On the other hand, we observe that the map

$$\Phi_1 : G_{ab} \otimes K_0(R) \to K_1(\mathbb{Z}[G])$$

from Corollary 3.9 sends the simple tensor $g \otimes [\mathbb{Z}] \in G_{ab} \otimes K_0(\mathbb{Z})$ to the element $\{\mu_{g^{-1}}\} \in K_1(\mathbb{Z}[G])$ represented by multiplication by g^{-1}. Property of group homomorphism then tells us that Φ_1 sends the simple tensor $g \otimes [\mathbb{Z}^n] \in G_{ab} \otimes K_0(\mathbb{Z})$ to the element $\{\mu_{g^{-n}}\} \in K_1(\mathbb{Z}[G])$ represented by multiplication by g^{-n}. As a result, under the identification

$$G_{ab} \otimes K_0(\mathbb{Z}) \cong G_{ab},$$

the map Φ_1 sends $g \in G_{ab}$ to $\{\mu_{g^{-1}}\} \in K_1(\mathbb{Z}[G])$.

Combining everything together, we see that, under the identification

$$\pi_1\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \cong \{1,-1\} \oplus G_{ab},$$

the Loday assembly $\pi_1(\alpha_{\mathrm{Loday}})$ sends the element $(\pm 1) \oplus g \in \{\pm 1\} \oplus G_{ab}$ to the element $\{\pm\mu_{g^{-1}}\} \in K_1(\mathbb{Z}[G])$. This proves $\mathrm{im}\left(\pi_1(\alpha_{\mathrm{Loday}})\right)$ is the subgroup $\pm G$ of $K_1(\mathbb{Z}[G])$ as desired. □

4. Applications to the Second K-Groups of Integral Group Rings

The following short exact sequence

$$0 \to K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \to \pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \to H_2(BG; K_0(\mathbb{Z})) \to 0 \tag{4.0.1}$$

is the extension problem of the Atiyah-Hirzebruch spectral sequence for computing the second homotopy group $\pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right)$. See [Leh18, Theorem 12.2 on page 29]. Together with Theorem 3.8, we give some injectivity results for the Loday assembly on the second K-group. We begin by the following corollary:

Corollary 4.1 (The Loday Assembly on π_2 of an Integral Group Ring) Let G be a group. The Loday assembly map

$$\alpha_{\mathrm{Loday}} : BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}} \to \mathbb{K}_{\mathbb{Z}[G]}^{\mathrm{GW}} \tag{4.0.2}$$

for the integral group ring $\mathbb{Z}[G]$ on π_2, when restricted onto the subgroup

$$K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})],$$

is given by the formula:

$$K_2(i) \oplus \Phi_2 : K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})] \to K_2(\mathbb{Z}[G]), \tag{4.0.3}$$

for which $i : \mathbb{Z} \to \mathbb{Z}[G]$ is the inclusion, and Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\to K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1, g\}_{\mathrm{St}}. \end{aligned} \tag{4.0.4}$$

In particular, the Steinberg symbol $\{1, g\}_{\mathrm{St}}$ is always the identity element in $K_2(\mathbb{Z}[G])$ for all $g \in G$.

Proof. Only the description of Φ_2 needs to be justified. Theorem 3.8 says Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\to K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1\} \star \{\mu_{g^{-1}}\}, \end{aligned} \tag{4.0.5}$$

and we have

$$\begin{aligned} \{\pm 1\} \star \{\mu_{g^{-1}}\} &= -\{\pm 1, \mu_{g^{-1}}\}_{\mathrm{St}} && \text{(Proposition 2.4)} \\ &= -\{\pm 1, g^{-1}\}_{\mathrm{St}} && \text{(By definition of } \mu_{g^{-1}}\text{)} \\ &= \left\{\pm 1, \left(g^{-1}\right)^{-1}\right\}_{\mathrm{St}} && \text{(Bilinearity of the Steinberg symbol)} \\ &= \{\pm 1, g\}_{\mathrm{St}} \end{aligned}$$

as desired. $\square$

We now see that the Steinberg symbol $\{-1, g\}_{\mathrm{St}}$ is closely related to the non-injectivity problem for the Loday assembly.

Question 4.2 (Non-injectivity Problem for the Loday Assembly)
Let R be a regular ring. Is there a group G with torsions, such that Loday assembly

$$\pi_i(\alpha_{\mathrm{Loday}}) : \pi_i\left(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \to K_i(R[G])$$

is not injective for some i?

Question 4.2 was answered by Ullmann-Wu in the case when R is a finite field.

Theorem 4.3 ([UW17, Theorem 2 on page 461])
Let G be a finite group such that $H_2(G;\mathbb{Z})$ is non-trivial, and $\mathbb{F}$ a finite field with characteristic p which does not divide the order of G, then the Loday assembly

$$\pi_2(\alpha_{\mathrm{Loday}}) : \pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{F}}^{\mathrm{GW}}\right) \to K_2(\mathbb{F}[G])$$

is not injective.

For example, we can take $G = C_2^{\oplus 2}$ and $\mathbb{F}$ to be any finite field with characteristic $p > 2$.

However, no such group is known when $R = \mathbb{Z}$. Note that under suitable conditions, the Loday assembly is injective in lower degrees.

Proposition 4.4 ([LR05, Lemma 2 on page 709])
Let R be a regular ring, and G be a group.

(i) The Loday assembly

$$\pi_0\left(\alpha_{\text{Loday}}\right) : \pi_0\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right) \to K_0(R[G])$$

on π_0 is always injective. In particular, its left inverse is induced by the augmentation map

$$R[G] \to R.$$

(ii) If in addition R is commutative, and the natural map

$$\begin{aligned} \mathbb{Z} &\to K_0(R) \\ 1 &\mapsto [R] \end{aligned}$$

is an isomorphism, then then Loday assembly

$$\pi_1\left(\alpha_{\text{Loday}}\right) : \pi_1\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right) \to K_1(R[G])$$

on π_1 is injective.

Consequently, the first place to look for non-injectivity phenomenon for the integral case of the Loday assembly would be the second homotopy group. The hope is that Corollary 4.1 will assist on answering the integral case of Question 4.2.

Example 4.5 (Injectivity on $K_2(\mathbb{Z}[C_2])$). Denote by C_2 the cyclic group of order two, with generator g. Dunwoody showed that the second K-group of $\mathbb{Z}[C_2]$ is generated by two Steinberg symbols [Dun75, Theorem on page 482]:

$$\begin{aligned} K_2\left(\mathbb{Z}[C_2]\right) &= \left\langle \{-1,-1\}_{\text{St}}, \{-1,g\}_{\text{St}} \right\rangle \\ &\cong C_2 \oplus C_2. \end{aligned} \tag{4.0.6}$$

Theorem 3.8 then says the Loday assembly

$$\alpha_{\text{Loday}} : (BC_2)_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}} \to \mathbb{K}_{\mathbb{Z}[C_2]}^{\text{GW}} \tag{4.0.7}$$

for the cyclic group of order 2 is injective on π_2. In fact, because $g = g^{-1}$ in C_2, we know it is bijective on π_2.

We have an expression for the Steinberg symbol $\{-1,g\}_{\text{St}}$ in terms of generators of the Steinberg group $\text{St}\left(\mathbb{Z}[G]\right)$.

Theorem 4.6
Let G be a group. The Steinberg symbol $\{-1, g\}_{\mathrm{St}} \in K_2(\mathbb{Z}[G])$ for $g \in G$ is given by

$$\{-1, g\}_{\mathrm{St}} = w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ w_{12}(g), \tag{4.0.8}$$

where

$$w_{ij}(u) := x_{ij}(u)\ x_{ji}(-u^{-1})\ x_{ij}(u), \tag{4.0.9}$$

and the $x_{ij}(u)$s' are the generators of the Steinberg group $\mathrm{St}\,(\mathbb{Z}[G])$.

Proof. The proof involves playing with the presentation of the Steinberg group.

We recall from [Ros95, Lemma 4.2.15 on page 195] the identities

$$w_{ij}(u)^{-1} = w_{ij}(-u), \tag{4.0.10}$$

$$w_{ij}(u) = w_{ji}(-u^{-1}), \tag{4.0.11}$$

$$w_{k\ell}(u)\ w_{ij}(v)\ w_{k\ell}(u)^{-1} = \begin{cases} w_{ij}(v) & \text{if } i,\ j,\ k,\ \ell \text{ are all distinct,} \\ w_{\ell j}(-u^{-1}v) & \text{if } k = i, \text{ and } i,\ j,\ \ell \text{ are all distinct,} \\ w_{i\ell}(-vu) & \text{if } k = j, \text{ and } i,\ j,\ k \text{ are all distinct,} \\ w_{ji}(-u^{-1}vu^{-1}) & \text{if } k = i, \text{ and } j = \ell. \end{cases} \tag{4.0.12}$$

In particular, Equation (4.0.12) gives

$$w_{13}(u)\ w_{12}(v) = w_{23}(v^{-1}u)\ w_{13}(u), \tag{4.0.13}$$

$$w_{13}(u)\ w_{23}(v) = w_{12}(-uv^{-1})\ w_{13}(u). \tag{4.0.14}$$

Put

$$h_{ij}(u) := w_{ij}(u)\ w_{ij}(-1), \tag{4.0.15}$$

then a long and boring computation shows

$$\begin{aligned}
\{-1, g\}_{\mathrm{St}} &:= [h_{12}(-1), h_{13}(g)] \\
&= h_{12}(-1)\ h_{13}(g)\ h_{12}(-1)^{-1}\ h_{13}(g)^{-1} \\
&= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{13}(-1)\ w_{12}(-1)^{-1}\ w_{12}(-1)^{-1}\ w_{13}(-1)^{-1}\ w_{13}(g)^{-1} \\
&= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{13}(-1)\ w_{12}(1)\ w_{12}(1)\ w_{13}(1)\ w_{13}(-g) \\
&\qquad (\text{Equation (4.0.10)}) \\
&= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ \underbrace{w_{13}(-1)\ w_{12}(1)}_{w_{23}(-1)\ w_{13}(-1)}\ w_{12}(1)\ w_{13}(1)\ w_{13}(-g)
\end{aligned}$$

(Equation (4.0.13))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{23}(-1)\ \underbrace{w_{13}(-1)\ w_{12}(1)}_{w_{23}(-1)\ w_{13}(-1)}\ w_{13}(1)\ w_{13}(-g)$$

(Equation (4.0.13))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{23}(-1)\ w_{23}(-1)\ \underbrace{w_{13}(-1)\ w_{13}(1)}_{1}\ w_{13}(-g)$$

(Equation (4.0.10))

$$= w_{12}(-1)\ w_{12}(-1)\ \underbrace{w_{13}(g)\ w_{23}(-1)}_{w_{12}(g)\ w_{13}(g)}\ w_{23}(-1)\ w_{13}(-g)$$

(Equation (4.0.14))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{13}(g)\ w_{23}(-1)\ w_{13}(-g)}_{w_{31}(-g^{-1})\ w_{23}(-1)\ w_{31}(g^{-1})}$$

(Equation (4.0.11))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{31}(-g^{-1})\ w_{23}(-1)\ w_{31}(g^{-1})}_{w_{21}(-g^{-1})}$$

(Equation (4.0.12))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{21}(-g^{-1})}_{w_{12}(g)}$$

(Equation (4.0.11))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ w_{12}(g)$$

as desired. □

The integral case of Question 4.2 on π_2 is then related to the word problem for the Steinberg group.

Theorem 4.7
Let G be a group such that the second homology group $H_2(BG;\mathbb{Z})$ is trivial. Then the Loday assembly

$$\pi_2\left(\alpha_{\mathrm{Loday}}\right) : \pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \to K_i(\mathbb{Z}[G])$$

on π_2 is non-injective if and only if there exists a non-identity element $g \in G$ such that the Steinberg symbol $\left\{-1, g^{-1}\right\}_{\mathrm{St}} \in K_2(\mathbb{Z}[G])$ is non-trivial.

Proof. If $H_2(BG;\mathbb{Z})$ is trivial, then the short exact sequence in Equation (4.0.1) says

$$\pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}) \cong K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})]\,.$$

Corollary 4.1 then tells us that the Loday assembly on π_2 is given by

$$\pi_2\left(\alpha_{\mathrm{Loday}}\right) = K_2(i) \oplus \Phi_2.$$

Now, the map $K_2(i)$ is always injective, so the injectivity/non-injectivity behaviour of $\pi_2(\alpha_{\mathrm{Loday}})$ is completely determined by Φ_2 in this case. Again, Corollary 4.1 says Φ_2 is non-injective if and only if there exists a non-identity element $g \in G$ such that the Steinberg symbol $\left\{-1, g^{-1}\right\}_{\mathrm{St}} \in K_2(\mathbb{Z}[G])$ is non-trivial. □

Example 4.8 (Kähler Differentials and de Rham Cohomology). Let k be a unital, commutative ring, and A be a unital, commutative k-algebra. We define the **A-module of Kähler differentials** to be the free A-module generated by the symbols da for each $a \in A$, modulo linearity and the Leibniz rule:

$$\Omega^1_{A|k} := \frac{\langle da \mid a \in A\rangle}{\left\langle \begin{array}{rl} d(\lambda a + \mu b) & = \lambda da + \mu db, \\ d(ab) & = a(db) + b(da) \end{array}\right\rangle}, \tag{4.0.16}$$

where λ, $\mu \in k$ and a, $b \in A$. The Leibniz rule implies $d(1_k) = 0$, and consequently, $du = 0$ for any $u \in k$.

We then define the module $\Omega^n_{A|k}$ of **differential n-forms** to be the exterior product

$$\Omega^n_{A|k} := \Lambda^n_A \Omega^1_{A|k}\,, \tag{4.0.17}$$

where the exterior product is over A, not k. It is spanned by the elements $a_0 da_1 \wedge \cdots \wedge da_n$, for $a_i \in A$, that we usually write $a_0 da_1 \cdots da_n$.

Let us put $\Omega^0_{A|k} := A$. Then for each $n \geq 0$, the **exterior differential operator** is defined by

$$\begin{aligned} d : \Omega^n_{A|k} &\to \Omega^{n+1}_{A|k} \\ a_0 da_1 \cdots da_n &\mapsto da_0 da_1 \cdots da_n. \end{aligned} \tag{4.0.18}$$

Since $d(1_k) = 0$, it follows that $d \circ d = 0$, and the following sequence

$$A := \Omega^0_{A|k} \xrightarrow{d} \Omega^1_{A|k} \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n_{A|k} \xrightarrow{d} \cdots \tag{4.0.19}$$

is a chain complex, called the **de Rham chain complex of A over k**. The homology groups of the de Rham chain complex are denoted as $H^n_{\mathrm{dR}}\left(A|k\right)$, called the **de Rham cohomology groups of A over k**.

Using the theory of cyclic homology and Chern character, Loday constructed a map

$$K_2(A) \to H^2_{\mathrm{dR}}\left(A|k\right) \tag{4.0.20}$$

that sends the Steinberg symbol $\{x, y\}_{\mathrm{St}}$ to the cohomology class represented by the differential form $x^{-1}y^{-1}dxdy$ [Lod97, page 275]. One may hope that this map will be useful to determine the triviality of the Steinberg symbol $\{-1, g\}_{\mathrm{St}}$ when $A = \mathbb{Z}[G]$ is an integral group ring. However, because $d(1) = 0$, the image of the Steinberg symbol $\{-1, g\}_{\mathrm{St}}$ is always trivial in $H^2_{\mathrm{dR}}(\mathbb{Z}[G]|\mathbb{Z})$. This means we need other methods to analyse $\{-1, g\}_{\mathrm{St}}$.

Theorem 4.6 motivates the following definition:

Definition 4.9 (The group $W_{12}(G)$)
Let G be a group. The subgroup $W_{12}(G)$ of $K_2(\mathbb{Z}[G])$ is generated by the Steinberg symbols:

$$W_{12}(G) := \langle \{-1, g\}_{\mathrm{St}} \mid g \in G \rangle . \tag{4.0.21}$$

Notice the following:

(1) The image of the map

$$\Phi_2 : G \times K_1(\mathbb{Z}) \to K_2(\mathbb{Z}[G])$$

in Equation (4.0.3) is precisely $W_{12}(G)$.

(2) Readers who are familiar with the classical definition

$$\mathrm{Wh}_2(G) := \frac{K_2(\mathbb{Z}[G])}{K_2(\mathbb{Z}[G]) \cap W(G)} \tag{4.0.22}$$

of the second Whitehead group proposed by Hatcher-Wagoner in [HW73, page 10] will see our group $W_{12}(G)$ follows their spirit. The group $W(G)$ is generated by the elements $w_{ij}(\pm g)$ for all $g \in G$, and for all i, j. Together with Theorem 4.6, we see that $W_{12}(G)$ is a subgroup of $W(G)$.

(3) From the functoriality of the Steinberg group, we get a functor

$$W_{12}(-) : \mathcal{G}\mathrm{roups} \to \mathcal{A}\mathrm{belian} . \tag{4.0.23}$$

Moreover, for every group G, there is a natural group homomorphism

$$\begin{aligned} \Psi_G : G &\to W_{12}(G) \\ g &\mapsto \{-1, g\}_{\mathrm{St}} , \end{aligned} \tag{4.0.24}$$

or equivalently the restriction

$$\Psi_G = \Phi_2|_{\{-1\} \times G} \tag{4.0.25}$$

for Φ_2 as in Equation (4.0.3). Non-injectivity of Ψ_G will imply the non-injectivity of the Loday assembly on π_2.

Example 4.10 (Injectivity on $K_2\left(\mathbb{Z}\left[C_2^{\oplus n}\right]\right)$). This example is due to Daniel Ramras.

Denote by C_2 the cyclic group of order 2. Fix $n \in \mathbb{N}$, and consider the n-fold direct sum. If $x \in C_2^{\oplus n}$ is non-zero, then there exists $1 \leq i \leq n$ such that the image $\mathrm{proj}_i(x)$ of x under the i-th projection map

$$\mathrm{proj}_i : C_2^{\oplus n} \to C_2$$

is non-zero. The functoriality of $W_{12}(-)$ then gives a commutative diagram

$$\begin{tikzcd}
C_2^{\oplus n} \arrow[r, "\Psi_{C_2^{\oplus n}}"] \arrow[d, "\mathrm{proj}_i"'] & W_{12}(C_2^{\oplus n}) \arrow[d, "W_{12}(\mathrm{proj}_i)"] \\
C_2 \arrow[r, "\Psi_{C_2}"'] & W_{12}(C_2).
\end{tikzcd} \tag{4.0.26}$$

We saw in Example 4.5 that Ψ_{C_2} is injective, so the commutativity of Diagram (4.0.26) says $\Psi_{C_2^{\oplus n}}(x) \in W_{12}(C_2^{\oplus n})$ is non-zero.

Let us recap what we did here. Fix a non-zero element $x \in C_2^{\oplus n}$. There exists $1 \leq i \leq n$ such that $\mathrm{proj}_i(x) \in C_2$ is non-zero. Because Ψ_{C_2} is injective, we must have

$$\left(W_{12}(\mathrm{proj}_i) \circ \Psi_{C_2^{\oplus n}}\right)(x) \neq 0,$$

and consequently $\Psi_{C_2^{\oplus n}}(x) \neq 0$, proving the group homomorphism

$$\Psi_{C_2^{\oplus n}} : C_2^{\oplus n} \to W_{12}(C_2^{\oplus n})$$

is injective for any $n \in \mathbb{N}$.

As a result, the Loday assembly map

$$\alpha_{\mathrm{Loday}} : B\left(C_2^{\oplus n}\right)_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}} \to \mathbb{K}_{\mathbb{Z}\left[C_2^{\oplus n}\right]}^{\mathrm{GW}}$$

for the group $C_2^{\oplus n}$ is injective on π_2 when restricted onto the subgroup $K_2(\mathbb{Z}) \oplus \left[C_2^{\oplus n} \otimes K_1(\mathbb{Z})\right]$.

Example 4.11 (Cyclic Groups of Odd Orders). Our formula does not work well on studying the K-theory of the integral group ring $\mathbb{Z}[C_n]$ of cyclic group C_n of odd order n. Since $C_n \otimes C_2 = 0$, the map Φ_2 in Equation (4.0.4) is the zero map. Because $H_2(BC_n; \mathbb{Z}) = 0$ for all odd number n, the short exact sequence in Equation (4.0.1) gives

$$\pi_2\left(BC_{n+} \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \cong K_2(\mathbb{Z}), \tag{4.0.27}$$

and therefore we learn nothing about structures of $K_2(\mathbb{Z}[C_n])$ from this approach, other than it contains $K_2(\mathbb{Z})$ as a subgroup.

5. Comparing Assemblies

In this section, we show our definition of Loday assembly can be used to compute the Weiss-Williams assembly (i.e., the universal assembly). We begin by recalling the setting of the Weiss-Williams assembly as in [WW95].

5.1. **Two Intermediate Spectra.**

5.1.1. *Symmetric Monoidal Categories and the $S^{-1}S$-Construction.*

Definition 5.1 (Symmetric Monoidal Category)
A **symmetric monoidal category** is a category $\mathcal{C}$ that has
(SMC 1) a functor

$$\begin{aligned} \Box : \mathcal{C} \times \mathcal{C} &\to \mathcal{C} \\ (A, B) &\mapsto A\Box B, \end{aligned}$$

(SMC 2) a distinguished object $e \in \mathcal{C}$,
(SMC 3) and four basic natural isomorphisms

$$\begin{aligned} e\Box A &\cong A, \\ A\Box e &\cong A, \\ (A\Box B)\Box C &\cong A\Box(B\Box C), \\ A\Box B &\cong B\Box A. \end{aligned}$$

that are subject to the coherence conditions given in [ML98].
A **monoidal functor** between symmetric monoidal categories is a functor that respects all the axioms. The category of symmetric monoidal categories and monoidal functors will be denoted as $\mathcal{S}\mathrm{ym}\mathcal{M}\mathrm{on}\mathcal{C}\mathrm{at}$.

Example 5.2 . The category $\mathcal{F}\mathrm{in}\mathcal{S}\mathrm{et}$ of finite sets is a symmetric monoidal category under disjoint union. The distinguished object is the empty set.

Example 5.3 . If $(\mathcal{C}, \Box)$ is a symmetric monoidal category, then its category $\mathrm{iso}\,(\mathcal{C})$ of isomorphisms is also a symmetric monoidal category under $\Box$.

The axioms (SMC 1)–(SMC 3) make the set $\pi_0(B\mathcal{C})$ of the path-connected components of the classifying space of the symmetric monoidal category $\mathcal{C}$ a monoid. The $S^{-1}S$-construction is a categorical method to study the group completion of this monoidal.

Definition 5.4 (The Category $S^{-1}S$, [Wei13, Definition 4.2 on page 328])
Let $(S, \Box)$ be a symmetric monoidal category. We define a new category $S^{-1}S$ from S by the following data:
(SIS 1) Objects in $S^{-1}S$ are pairs (m, n) of objects in S.
(SIS 2) A morphism $(m_1, m_2) \to (n_1, n_2)$ in $S^{-1}S$ is an equivalence class of composite

$$(m_1, m_2) \xrightarrow{s\Box} (s\Box m_1, s\Box m_2) \xrightarrow{(f,g)} (n_1, n_2).$$

This composite is equivalent to

$$(m_1, m_2) \xrightarrow{t\square} (t\square m_1, t\square m_2) \xrightarrow{(f', g')} (n_1, n_2)$$

if and only if there is an isomorphism $\alpha : s \to t$ in S so that the composition with $\alpha\square m_i$ sends f' and g' to f and g.

We note that a monoidal functor $S \to T$ induces a functor

$$S^{-1}S \to T^{-1}T.$$

Moreover, the category $S^{-1}S$ has a natural symmetric monoidal structure induced by the symmetric monoidal structure on S (see [Wei13, Remark 4.2.2 on page 329]). Thus the $S^{-1}S$-construction gives a functor

$$\mathcal{S}\mathrm{ym}\mathcal{M}\mathrm{on}\mathcal{C}\mathrm{at} \to \mathcal{S}\mathrm{ym}\mathcal{M}\mathrm{on}\mathcal{C}\mathrm{at}$$
$$S \mapsto S^{-1}S.$$

In particular, this symmetric monoidal structure makes $\pi_0\left(BS^{-1}S\right)$ an Abelian group. Under the conditions as in [Wei13, Theorem 4.8 on page 333], $\pi_0\left(BS^{-1}S\right)$ is a group completion of $\pi_0(BS)$.

Definition 5.5 (The Category $\mathcal{S}_{\mathcal{A}}$)
Let $\mathcal{A}$ be a symmetric monoidal category. We define the category $\mathcal{S}_{\mathcal{A}}$ to be the $S^{-1}S$-construction of the category $\mathrm{iso}(\mathcal{A})$ of isomorphisms in $\mathcal{A}$, as defined in Definition 5.4:

$$\mathcal{S}_{\mathcal{A}} := \mathrm{iso}(\mathcal{A})^{-1}\,\mathrm{iso}(\mathcal{A}) \tag{5.1.1}$$

In particular, we write

$$\mathcal{S}_R := \mathcal{S}_{\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}} \tag{5.1.2}$$

for a ring R.

We say a ring R satisfies the **Invariant Basis Property (IBP)** when $R^n \cong R^m$ if and only if $m = n$. Under the IBP condition, we have the following description of the group completion for the classifying space $B\,\mathrm{iso}\left(\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}\right)$.

Theorem 5.6 ([Wei13, Theorem 4.9 on page 334])
Let R be a ring satisfying the IBP. Then the classifying space of the category $\mathrm{iso}\left(\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}\right)$ is given by

$$B\,\mathrm{iso}\left(\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}\right) = \bigsqcup_{p\geq 1} BGL(p, R). \tag{5.1.3}$$

Moreover, the space $\mathbb{Z} \times BGL(R)^+$ is a model for the classifying space of the $S^{-1}S$-construction of the category $\mathrm{iso}\left(\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}\right)$:

$$B\mathcal{S}_R \simeq \mathbb{Z} \times BGL(R)^+, \tag{5.1.4}$$

with the natural group completion map

$$\mathfrak{gc}_R : \coprod_{p\geq 1} BGL(p,R) \to \mathbb{Z} \times BGL(R)^+ \tag{5.1.5}$$

given by sending a matrix $A_p \in GL(p,R)$ to $(p, A_p) \in \mathbb{Z} \times GL(R)$.

5.1.2. *A Categorical Description of the Loday Pairing and the Spectrum $\mathbb{K}_R^{\mathrm{free}}$.*

Under Theorem 5.6, we give another formalism of the Loday pairing γ_{Loday} define in Definition 2.2 using category theory. This formalism is well-documented in [Wei81] and we shall review it below.

Again, let us choose an isomorphism $R^m \otimes S^n \cong (R \otimes S)^{mn}$. Tensor product of modules gives a functor

$$\otimes : \mathrm{iso}\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \times \mathrm{iso}\left(\mathfrak{Free}_S^{\mathrm{fg}}\right) \to \mathrm{iso}\left(\mathfrak{Free}_{R\otimes S}^{\mathrm{fg}}\right). \tag{5.1.6}$$

We then have the following homotopy commutative diagram

$$\begin{tikzcd}
B\,\mathrm{iso}\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \wedge B\,\mathrm{iso}\left(\mathfrak{Free}_S^{\mathrm{fg}}\right) \arrow[r, "B\otimes"] \arrow[d, equal] & B\,\mathrm{iso}\left(\mathfrak{Free}_{R\otimes S}^{\mathrm{fg}}\right) \arrow[d, equal] \\
\left(\coprod_{p,q\geq 1} BGL(p,R) \times BGL(q,S)\right)_+ \arrow[r, "B\otimes"] \arrow[d] & \coprod_{r\geq 0} BGL(r, R\otimes S) \arrow[d] \\
\left(\coprod_{p,q\geq 1} BGL(p,R)^+ \times BGL(q,S)^+\right)_+ \arrow[r, "f"] \arrow[d, equal] & \coprod_{r\geq 0} BGL(r, R\otimes S)^+ \arrow[dd, "\mathfrak{gc}_{R\otimes S}"] \\
\left(\coprod_{p\geq 0} BGL(p,R)^+\right) \wedge \left(\coprod_{q\geq 0} BGL(q,S)^+\right) \arrow[d, "\mathfrak{gc}_R \wedge \mathfrak{gc}_S"'] & \\
\left(\mathbb{Z} \times BGL(R)^+\right) \wedge \left(\mathbb{Z} \times BGL(S)^+\right) \arrow[r, dashed, "\gamma_{\mathrm{free}}"'] & \mathbb{Z} \times BGL(R\otimes S)^+.
\end{tikzcd} \tag{5.1.7}$$

The map f induced by $B\otimes$ via the functoriality of the $+$-construction.

The group completion $\mathbb{Z} \times BGL(R)^+$ satisfies the following **cofinality condition**:

Condition 5.7 (Confinality Condition)
For every $x := (\xi, \mathcal{M}) \in \mathbb{Z} \times BGL(R)^+$, there exists $m \in \mathbb{N}$ and $x_0 := M_p \in BGL(p,R)^+$ such that

$$(\xi, \mathcal{M}) + (m, I_m) = \mathfrak{gc}_R(M_p).$$

We want to define a map

$$\gamma_{\text{free}} : \left(\mathbb{Z} \times BGL(R)^+\right) \wedge \left(\mathbb{Z} \times BGL(S)^+\right) \to \mathbb{Z} \times BGL(R \otimes S)^+$$

so that Diagram (5.1.7) commutes up to weak homotopy.

Definition 5.8 (The map $\widetilde{\gamma_{\text{free}}}$)
Under the notations in the confinality Condition (5.7), define a map

$$\widetilde{\gamma_{\text{free}}} : \left(\mathbb{Z} \times BGL(R)^+\right) \times \left(\mathbb{Z} \times BGL(S)^+\right) \to \mathbb{Z} \times BGL(R \otimes S)^+$$

by

$$\widetilde{\gamma_{\text{free}}}(x, y) := \mathfrak{gc}_{R\otimes S} \circ f(x_0, y_0) - \mathfrak{gc}_{R\otimes S} \circ f(*_m, y_0) - \mathfrak{gc}_{R\otimes S} \circ f(x_0, *_n) + \mathfrak{gc}_{R\otimes S} \circ f(*_m, *_n)\,.$$

Here, $*_m$ (resp. $*_n$) is the base-point of $BGL(m, R)^+$ (resp. $BGL(n, S)^+$). The plus and minus are from the H-group structure of $\mathbb{Z} \times BGL(R \otimes S)^+$.

We leave readers to verify the map $\widetilde{\gamma_{\text{free}}}$ is well-defined.

Lemma 5.9
Under the notations in Diagram (5.1.7), the following diagram

$$\begin{tikzcd}
\left(\coprod_{p\geq 0} BGL(p,R)^+\right) \times \left(\coprod_{q\geq 0} BGL(q,S)^+\right) \arrow[r, "f"] \arrow[d, "\mathfrak{gc}_R \wedge \mathfrak{gc}_S"'] & \coprod_{r\geq 0} BGL(r, R\otimes S)^+ \arrow[d, "\mathfrak{gc}_{R\otimes S}"] \\
\left(\mathbb{Z} \times BGL(R)^+\right) \times \left(\mathbb{Z} \times BGL(S)^+\right) \arrow[r, "\widetilde{\gamma_{\text{free}}}"'] & \mathbb{Z} \times BGL(R\otimes S)^+.
\end{tikzcd}$$

commutes up to homotopy.

Proof. This is just a matter of diagram-chasing. Consider a point x in the image of $\mathfrak{gc}_R$. We know

$$x = (p, M_p)$$

for some $M_p \in BGL(p, R)^+$. Confinality condition (5.7) says there exists $m \in \mathbb{N}$ such that

$$(p, M_p) + (m, I_m) = (p + m, M_p \boxplus I_m)$$

is in the image of $\mathfrak{gc}_R$, where $\boxplus$ denotes the block sum of matrices. In fact, we have

$$(p + m, M_p \boxplus I_m) = \mathfrak{gc}_R(M_p \boxplus I_m)\,.$$

Similarly, for $y = (q, N_q) \in \mathbb{Z} \times BGL(S)^+$, there exists $n \in \mathbb{N}$ such that

$$(q + n, N_q \boxplus I_n) = \mathfrak{gc}_S(N_q \boxplus I_n)\,.$$

We then compute

$$
\begin{aligned}
\widetilde{\gamma_{\text{free}}}(x,y) &= \widetilde{\gamma_{\text{free}}}((p,M_p),(q,N_q)) \\
&:= \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, N_q \boxplus I_n) - \mathfrak{gc}_{R\otimes S} \circ f(*_m, N_q \boxplus I_n) \\
&\quad - \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, *_n) + \mathfrak{gc}_{R\otimes S} \circ f(*_m, *_n) \\
&= \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, N_q \boxplus I_n) - \mathfrak{gc}_{R\otimes S} \circ f(I_m, N_q \boxplus I_n) \\
&\quad - \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, I_n) + \mathfrak{gc}_{R\otimes S} \circ f(I_m, I_n) \\
&\sim \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, N_q \boxplus I_n) - \mathfrak{gc}_{R\otimes S} \circ f(N_q \boxplus I_n, I_m) \\
&\quad - \mathfrak{gc}_{R\otimes S} \circ f(M_p \boxplus I_m, I_n) + \mathfrak{gc}_{R\otimes S} \circ f(I_m, I_n) \\
&= \mathfrak{gc}_{R\otimes S}\left(M_p \otimes N_q \boxplus M_p^{\boxplus n} \boxplus N_q^{\boxplus m} \boxplus I_{mn}\right) - \mathfrak{gc}_{R\otimes S}\left(N_q^{\boxplus m} \boxplus I_{mn}\right) \\
&\quad - \mathfrak{gc}_{R\otimes S}\left(M_p^{\boxplus n} \boxplus I_{mn}\right) - \mathfrak{gc}_{R\otimes S}(I_{mn}) \\
&= (pq, M_p \otimes N_q) \\
&= \mathfrak{gc}_{R\otimes S}(M_p \otimes N_q)
\end{aligned}
$$

as desired. □

From the definition, it is clear that $\widetilde{\gamma_{\text{free}}}$ is homotopically trivial on the wedge

$$\left(\mathbb{Z} \times BGL(R)^+\right) \vee \left(\mathbb{Z} \times BGL(S)^+\right),$$

hence $\widetilde{\gamma_{\text{free}}}$ factors through the smash product to give the map

$$\gamma_{\text{free}} : \left(\mathbb{Z} \times BGL(R)^+\right) \wedge \left(\mathbb{Z} \times BGL(S)^+\right) \to \mathbb{Z} \times BGL(R \otimes S)^+, \tag{5.1.8}$$

making Diagram (5.1.7) commutes up to weak homotopy. In particular, when restricting onto the base-point component $\left(\{0\} \times BGL(R)^+\right) \wedge \left(\{0\} \times BGL(S)^+\right)$, we have

$$\gamma_{\text{free}}|_{\left(\{0\}\times BGL(R)^+\right)\wedge\left(\{0\}\times BGL(S)^+\right)} = \gamma_{\text{Loday}} \tag{5.1.9}$$

We are now able to define an intermediate spectrum that allows us to compare the Loday assembly and the universal assembly. Let us mimic the construction given in Proposition 2.12 and recall the map

$$\mathfrak{t}^+ : B\langle t \rangle \to K^+_{\Sigma\mathbb{Z}}$$

defined in Definition 2.11.

Definition 5.10 (The Spectrum $\mathbb{K}_R^{\text{free}}$)
Let R be a ring satisfying the IBP. Define the spectrum $\mathbb{K}_R^{\text{free}}$ by having

$$\left(\mathbb{K}_R^{\text{free}}\right)_n := \mathbb{Z} \times BGL(\Sigma^n R)^+ \\ \simeq B\mathcal{S}_{\Sigma^n R} \tag{5.1.10}$$

for $n \geq 0$. Recall our notation that

$$K_R^+ := K_0(R) \times BGL(R)^+ .$$

We write

$$K_R^{\text{free}} := \mathbb{Z} \times BGL(R)^+ \tag{5.1.11}$$

The structure maps are given by the composition

$$B\langle t\rangle \wedge K_{\Sigma^n R}^{\text{free}} \xrightarrow{\mathrm{t}^+ \wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^{\text{free}} \wedge K_{\Sigma^n R}^{\text{free}} \xrightarrow{\gamma_{\text{free}}} K_{\Sigma^{n+1} R}^{\text{free}}. \tag{5.1.12}$$

It is worth pointing out that

$$\pi_i(\mathbb{K}_R^{\text{free}}) \cong K_i(R)$$

for all $i \geq 0$. However, $\mathbb{K}_R^{\text{free}}$ is not an Ω-spectrum, unlike the Gersten-Wagoner spectrum $\mathbb{K}_R^{\text{GW}}$. So $\mathbb{K}_R^{\text{free}}$ and $\mathbb{K}_R^{\text{GW}}$ are two non-equivalent spectra. At last but not least, because of the construction of the structure maps in Equation (5.1.12), the pairing γ_{free} constructed in Equation (5.1.8) extends to a pairing of spectra. By abuse of notation, we shall denote it as

$$\gamma_{\text{free}} : \mathbb{K}_R^{\text{free}} \wedge \mathbb{K}_S^{\text{free}} \to \mathbb{K}_{R\otimes S}^{\text{free}} \tag{5.1.13}$$

for any two rings R, S satisfying the IBP.

5.1.3. *Idempotent Completion and the Spectrum* $\mathbb{K}_R^{\text{proj}}$.

A quick inspection of the classifying space $B\mathcal{S}_R$ given in Theorem 5.6 says

$$\pi_0\left(B\mathcal{S}_R\right) \ncong K_0(R).$$

The spectrum $\mathbb{K}_R^{\text{free}}$ defined in Definition 5.10 fixes this problem at the spectrum-level, and we provide a space-level solution to this problem here.

Definition 5.11 (Idempotent Completion, [Wei13, page 143])
The **idempotent completion** of a category $\mathcal{C}$ is the category $\widehat{\mathcal{C}}$ whose objects are pairs (C, p) with $p : C \to C$ an idempotent endomorphism of an object C of $\mathcal{C}$.

A morphism $(C, p) \to (C', p')$ in $\widehat{\mathcal{C}}$ is a map $f : C \to C'$ in $\mathcal{C}$ such that the diagram

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "p"'] & C' \arrow[d, "p'"] \\ C \arrow[r, "f"'] & C' \end{tikzcd} \tag{5.1.14}$$

commutes.

Example 5.12 . Because any projective module is a direct summand of a free module, we have

$$\widehat{\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}} \simeq \mathcal{P}\mathrm{roj}_R^{\mathrm{fg}}. \tag{5.1.15}$$

If $\mathcal{A}$ is a symmetric monoidal category, then so is $\widehat{\mathcal{A}}$. Thus, we define:

Definition 5.13 (The Category $\mathcal{P}_{\mathcal{A}}$)
Let $\mathcal{A}$ be a symmetric monoidal category. We define the category $\mathcal{P}_{\mathcal{A}}$ to be the $S^{-1}S$-construction of the category $\mathrm{iso}\left(\widehat{\mathcal{A}}\right)$ of isomorphisms in the idempotent completion $\widehat{\mathcal{A}}$, as defined in Definition 5.4:

$$\mathcal{P}_{\mathcal{A}} := \mathrm{iso}\left(\widehat{\mathcal{A}}\right)^{-1}\mathrm{iso}\left(\widehat{\mathcal{A}}\right) \tag{5.1.16}$$

In other words, $\mathcal{P}_{\mathcal{A}} := \mathcal{S}_{\widehat{\mathcal{A}}}$.
In particular, we write

$$\mathcal{P}_R := \mathcal{P}_{\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}} \tag{5.1.17}$$

for a ring R.

The following theorem tells us how $\mathcal{S}_R$ and $\mathcal{P}_R$ are related.

Theorem 5.14 ([CP97])
If $\mathcal{C}$ is a category, then there is a canonical embedding

$$\begin{aligned} \mathcal{C} &\to \widehat{\mathcal{C}} \\ C &\mapsto (C, \mathrm{id}). \end{aligned} \tag{5.1.18}$$

Therefore, we have an induced functor

$$\mathcal{S}_{\mathcal{C}} \to \mathcal{P}_{\mathcal{C}} \tag{5.1.19}$$

between the categories defined in Definition 5.5 and Definition 5.13.
When $\mathcal{C} = \mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}$ is the category of finitely generated free modules over the ring R, the induced group homomorphism

$$\pi_n\left(B\mathcal{S}_R\right) \to \pi_n\left(B\mathcal{P}_R\right) \tag{5.1.20}$$

is an isomorphism for $n \geq 1$.

Just as how we define the spectrum $\mathbb{K}_R^{\mathrm{free}}$, we need a correct formalism of the pairing.

Definition 5.15 (Pairing of Symmetric Monoidal Categories, [May80, page 308])
Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be symmetric monoidal categories. A **pairing of symmetric monoidal categories** is a functor

$$\otimes : \mathcal{A} \times \mathcal{B} \to \mathcal{C} \tag{5.1.21}$$

$$(A, B) \mapsto A \otimes B$$

satisfying the following condition:

(SMC 1) For any objects $A \in \mathcal{A}$, $B \in \mathcal{B}$, we have

$$0_{\mathcal{A}} \otimes B = 0_{\mathcal{C}} = A \otimes 0_{\mathcal{B}} \tag{5.1.22}$$

(SMC 2) For any objects A, $A' \in \mathcal{A}$ and B, $B' \in \mathcal{B}$, there is a coherent natural bi-distributivity isomorphism:

$$(A \square_{\mathcal{A}} A') \otimes (B \square_{\mathcal{B}} B') \cong (A \otimes B) \square_{\mathcal{C}} (A \otimes B') \square_{\mathcal{C}} (A' \otimes B) \square_{\mathcal{C}} (A' \otimes B') \,. \tag{5.1.23}$$

Example 5.16 . The standard example of a pairing of symmetric monoidal categories is the tensor product of modules.

Example 5.17 . If $\otimes : \mathcal{A} \times \mathcal{B} \to \mathcal{B}$ is a pairing of symmetric monoidal categories, then the category $\mathrm{iso}\,(\mathcal{A})$ of isomorphisms in $\mathcal{A}$ is also a symmetric monoidal category, and the pairing $\otimes$ induces a pairing

$$\otimes : \mathrm{iso}\,(\mathcal{A}) \times \mathrm{iso}\,(\mathcal{B}) \to \mathrm{iso}\,(\mathcal{C})$$

of symmetric monoidal categories.

Theorem 5.18 ([May80, Theorem 1.6 on page 307, and Theorem 2.1 on page 310])
A pairing $\mathcal{A} \times \mathcal{B} \to \mathcal{C}$ of symmetric monoidal categories functorially determines a pairing

$$B\mathcal{S}_{\mathcal{A}} \wedge B\mathcal{S}_{\mathcal{B}} \to B\mathcal{S}_{\mathcal{C}} \tag{5.1.24}$$

of topological spaces. Moreover, this also determines a pairing of their associated Ω-spectra.

Example 5.19 . In light of Theorem 5.6 and Lemma 5.9, the map

$$\gamma_{\mathrm{free}} : K_R^{\mathrm{free}} \wedge K_S^{\mathrm{free}} \to K_{R \otimes S}^{\mathrm{free}}$$

is the pairing functorially determined by the tensor product

$$\otimes : \mathrm{iso}\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \times \mathrm{iso}\left(\mathfrak{Free}_S^{\mathrm{fg}}\right) \to \mathrm{iso}\left(\mathfrak{Free}_{R \otimes S}^{\mathrm{fg}}\right)$$

of modules.

Definition 5.20 (The Pairing γ_{proj})
Let R, S be rings.
We define the pairing

$$\gamma_{\mathrm{proj}} : B\mathcal{P}_R \wedge B\mathcal{P}_S \to B\mathcal{P}_{R\otimes S} \tag{5.1.25}$$

to be the pairing that is functorially determined by the tensor product

$$\otimes : \mathrm{iso}\left(\widehat{\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_R}\right) \times \mathrm{iso}\left(\widehat{\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_S}\right) \to \mathrm{iso}\left(\widehat{\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_{R\otimes S}}\right)$$

of modules in the sense of Theorem 5.18.

Because

$$\begin{aligned} B\mathcal{P}_R &\simeq K_0(R) \times BGL(R)^+ \\ &=: K_R^+ \end{aligned} \tag{5.1.26}$$

by [Wei13, Corollary 4.11.1 on page 337], the pairing γ_{proj} is actually a map

$$\gamma_{\mathrm{proj}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+. \tag{5.1.27}$$

If in addition that the rings R, S both satisfy the IBP, then functoriality of the pairings says the following diagram

$$\begin{tikzcd}
B\,\mathrm{iso}\left(\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_R\right) \wedge B\,\mathrm{iso}\left(\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_S\right) \arrow[rr, "B\otimes"] \arrow[dr, "B\,(\mathrm{inclusion})"] \arrow[dd] & & B\,\mathrm{iso}\left(\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_{R\otimes S}\right) \arrow[dr, "B\,(\mathrm{inclusion})"] \arrow[dd] & \\
& B\,\mathrm{iso}\left(\mathcal{P}\mathrm{roj}^{\mathrm{fg}}_R\right) \wedge B\,\mathrm{iso}\left(\mathcal{P}\mathrm{roj}^{\mathrm{fg}}_S\right) \arrow[rr, "B\otimes"] \arrow[dd] & & B\,\mathrm{iso}\left(\mathcal{P}\mathrm{roj}^{\mathrm{fg}}_{R\otimes S}\right) \arrow[dd] \\
B\mathcal{S}_R \wedge B\mathcal{S}_S \arrow[rr, "\gamma_{\mathrm{free}}"] \arrow[dr, "B(5.1.19)"'] & & B\mathcal{S}_{R\otimes S} \arrow[dr, "B(5.1.19)"'] & \\
& B\mathcal{P}_R \wedge B\mathcal{P}_S \arrow[rr, "\gamma_{\mathrm{proj}}"'] & & B\mathcal{P}_{R\otimes S}
\end{tikzcd} \tag{5.1.28}$$

commutes up to weak homotopy, where the vertical arrows are the group completion maps. Consequently, the two pairings γ'_{Loday} and γ_{proj} are not homotopic by Proposition 2.7.

Definition 5.21 (The Spectrum $\mathbb{K}_R^{\mathrm{proj}}$)
Let R be a ring. Define the spectrum $\mathbb{K}_R^{\mathrm{proj}}$ by having

$$\left(\mathbb{K}_R^{\mathrm{proj}}\right)_n := K_{\Sigma^n R}^+ \tag{5.1.29}$$

$$\simeq B\mathcal{P}_{\Sigma^n R}$$

for $n \geq 0$. The structure maps are given by the composition

$$B\langle t\rangle \wedge K_R^+ \xrightarrow{t^+ \wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \xrightarrow{\gamma_{\mathrm{proj}}} K_{\Sigma R}^+ \tag{5.1.30}$$

Again, the pairing γ_{proj} constructed in Equation (5.1.25) extends to a pairing of spectra. By abuse of notation, we shall denote it as

$$\gamma_{\mathrm{proj}} : \mathbb{K}_R^{\mathrm{proj}} \wedge \mathbb{K}_S^{\mathrm{proj}} \to \mathbb{K}_{R\otimes S}^{\mathrm{proj}} \tag{5.1.31}$$

for any two rings R, S.

The following result is an immediate consequence of Theorem 5.18, Definition 5.10 and Definition 5.21.

Theorem 5.22
Let R, S be rings satisfying the IBP.

The inclusion functor

$$\mathfrak{Free}_R^{\mathrm{fg}} \to \mathcal{P}\mathrm{roj}_R^{\mathrm{fg}}$$

induces the following diagram

$$\begin{tikzcd}
\mathbb{K}_R^{\mathrm{free}} \wedge \mathbb{K}_S^{\mathrm{free}} \arrow[r, "\gamma_{\mathrm{free}}"] \arrow[d] & \mathbb{K}_{R\otimes S}^{\mathrm{free}} \arrow[d] \\
\mathbb{K}_R^{\mathrm{proj}} \wedge \mathbb{K}_S^{\mathrm{proj}} \arrow[r, "\gamma_{\mathrm{proj}}"'] & \mathbb{K}_{R\otimes S}^{\mathrm{proj}}
\end{tikzcd} \tag{5.1.32}$$

of weak pairings of spectra. More precisely, for each pair (m, n) of non-negative integers, the diagram

$$\begin{tikzcd}
K_{\Sigma^m R}^{\mathrm{free}} \wedge K_{\Sigma^n S}^{\mathrm{free}} \arrow[r, "\gamma_{\mathrm{free}}"] \arrow[d] & K_{\Sigma^{m+n}(R\otimes S)}^{\mathrm{free}} \arrow[d] \\
K_{\Sigma^m R}^{+} \wedge K_{\Sigma^n S}^{+} \arrow[r, "\gamma_{\mathrm{proj}}"'] & K_{\Sigma^{m+n}(R\otimes S)}^{+}
\end{tikzcd} \tag{5.1.33}$$

commutes up to weak homotopy.

The presence of the pairing map γ_{free} allows one to define a multiplication map

$$\begin{aligned}
\star_{\mathrm{free}} : K_i(R) \otimes K_j(S) &\to K_{i+j}(R\otimes S) \\
[f]\otimes[g] &\mapsto [\gamma_{\mathrm{free}} \circ (f \wedge g)]
\end{aligned} \tag{5.1.34}$$

It is clear from Diagram (5.1.7) that $\star_{\mathrm{free}}$ recovers tensor product of modules when $i = j = 0$, and coincides with Loday's multiplication $\star$ by Equation (5.1.9) when $i, j \geq 1$. However, the following cases are still open:

Question 5.23
What is the product map

$$\star_{\text{free}} : K_0(R) \otimes K_j(S) \to K_j(R \otimes S)? \tag{5.1.35}$$

In particular, can we relate this product map with the classical product map

$$\# : K_0(R) \otimes K_1(S) \to K_1(R \otimes S)$$

defined by Milnor as in Equation (2.3.2) when $j = 1$?

5.2. **Categories Over Metric Spaces and the Non-Connective Pedersen-Weibel Algebraic K-Theory Spectrum.** Recall the definition of an additive category.

Definition 5.24 (Additive Category)
A category $\mathcal{A}$ is **additive** if all of the following are satisfied:

(A1) The collection of morphisms $\operatorname{Hom}(A, B)$ from A to B is an Abelian group for each object $A,\, B \in \mathcal{A}$.

(A2) The composition of morphisms

$$\operatorname{Hom}(A, B) \times \operatorname{Hom}(B, C) \to \operatorname{Hom}(A, C)$$

is a bilinear map.

(A3) There is a distinguished object $0_{\mathcal{A}} \in \mathcal{A}$ such that

$$\operatorname{Hom}(A, 0_{\mathcal{A}}) = 0 = \operatorname{Hom}(0_{\mathcal{A}}, A)$$

for all objects $A \in \mathcal{A}$.

(A4) There is a binary operation

$$\oplus : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$$

which is both the categorical product and categorical coproduct.

Example 5.25 . Our standard example is the category $\mathfrak{Free}_R^{\text{fg}}$ of finitely generated free modules over the ring R. The binary operation in (A4) is the direct sum of modules. Note that the ring R does not have to satisfy the IBP for $\mathfrak{Free}_R^{\text{fg}}$ being additive.

Definition 5.26 (Filtered Additive Category, [PW85, Definition 1.1 on page 168])
An additive category $\mathcal{A}$ is said to be **filtered** if there is an increasing filtration

$$F_0(A, B) \subseteq \cdots \subseteq F_n(A, B) \subseteq \cdots \subseteq \operatorname{Hom}(A, B)$$

on $\operatorname{Hom}(A, B)$ for every pair of objects $A,\, B \in \mathcal{A}$, satisfying the following properties:

(F1) Each $F_i(A, B)$ is a subgroup of $\operatorname{Hom}(A, B)$.

(F2) $\bigcup_i F_i(A,B) = \mathrm{Hom}\,(A,B)$.
(F3) The identity map id_A and the zero map 0_A are in $F_0(A,A)$.
(F4) The canonical maps

$$\begin{aligned} A \oplus B &\to A, \\ A &\to A \oplus B \end{aligned}$$

and all coherence isomorphisms are in $F_0(A,B)$.
(F5) The composition law in $\mathcal{A}$ respects the filtration. Meaning if $f \in F_i(A,B)$ and $g \in F_j(B,C)$, then $g \circ f \in F_{i+j}(A,C)$.

Definition 5.27 (The Category $\mathcal{C}_M(\mathcal{A})$, [Spe04, Definition 4.2 on page 14]) Let $\mathcal{A}$ be a category and (M,d) be a metric space. We definition the category $\mathcal{C}_M(\mathcal{A})$ by having

(i) objects are sets $\{A_x\}_{x\in M}$ of objects $A_x \in \mathcal{A}$, with the condition that the collection $\{x \in M \mid A_x \neq 0\}$ is locally finite.
(ii) A morphism $\varphi : A \to B$ in $\mathcal{C}_M(\mathcal{A})$ is a collection of morphisms

$$\varphi_y^x : A_x \to B_y$$

in $\mathcal{A}$ with a bound $R \geq 0$, so that $\varphi_y^x = 0$ whenever $d(x,y) > R$. In this case, we say the morphism $\varphi : A \to B$ is **bounded** by R.

Note that if $\mathcal{A}$ is an additive category, then so does $\mathcal{C}_M(\mathcal{A})$. Secondly, the boundedness condition on morphisms gives a natural filtration on the Hom-sets of $\mathcal{C}_M(\mathcal{A})$—for each pair of objects A, $B \in \mathcal{C}_M(\mathcal{A})$, the collection $F_i(A,B)$ consists of morphisms $A \to B$ in $\mathcal{C}_M(\mathcal{A})$ that are bounded by i. Thus the category $\mathcal{C}_M(\mathcal{A})$ is filtered additive in the sense of Definition 5.26. Thirdly, if the category $\mathcal{A}$ we start with is already filtered, we require all components $\varphi_y^x : A_x \to B_y$ of the morphism $\varphi : A \to B$ have filtration degree i as well.

Example 5.28 ($\mathcal{C}_i(\mathcal{A})$). Our standard example for $\mathcal{C}_M(\mathcal{A})$ is the case when $M = \mathbb{Z}^i$. Let us equip $\mathbb{Z}^i$ with the metric induced by the ℓ^∞-norm, so that

$$\begin{aligned} d(\vec{x},\vec{y}) &:= \|\vec{x}-\vec{y}\|_{\ell^\infty} \\ &:= \max_{1\leq j\leq i} |x_j - y_j| \end{aligned}$$

for $\vec{x} = (x_1, \cdots, x_i)$, $\vec{y} = (y_1, \cdots, y_i) \in \mathbb{Z}^i$. In this case, we write

$$\mathcal{C}_i(\mathcal{A}) := \begin{cases} \mathcal{A} & \text{if } i = 0, \\ \mathcal{C}_{\mathbb{Z}^i}(\mathcal{A}) & \text{if } i > 0 \end{cases} \tag{5.2.1}$$

and get a sequence $\{\mathcal{C}_i(\mathcal{A})\}_{i=0}^\infty$ of categories. Note that the categories $\mathcal{C}_i(\mathcal{C}_j(\mathcal{A}))$ and $\mathcal{C}_{i+j}(\mathcal{A})$ are isomorphic. The sequence $\{\mathcal{C}_i(\mathcal{A})\}_{i=0}^\infty$ was first constructed by Pedersen-Weibel to produce non-connective spectra from additive categories [PW85]. In particular, when working with the category of finitely generated free modules, the resulting spectrum has homotopy groups isomorphic to algebraic K-groups of the underlying ring. The categories $\mathcal{C}_M(\mathcal{A})$ were also studied by Carlsson-Pedersen to prove the K-theoretic Novikov Conjecture for a large class of groups [CP95].

Finally, we give $\mathcal{C}_i(\mathcal{A})$ the split exact structure as in [PW85, page 171]. More precisely, We say a chain

$$A \to C \to B$$

of morphisms in $\mathcal{C}_1(\mathcal{A})$ is an exact sequence if there is an isomorphism $A \oplus B \xrightarrow{\cong} C$ in $\mathcal{C}_1(\mathcal{A})$ such that the diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "\mathrm{id}"'] & A \oplus B \arrow[r] \arrow[d, "\cong"'] & B \arrow[d, "\mathrm{id}"] \\ A \arrow[r] & C \arrow[r] & B \end{tikzcd} \tag{5.2.2}$$

commutes. Because $\mathcal{C}_i(\mathcal{C}_j(\mathcal{A})) \cong \mathcal{C}_{i+j}(\mathcal{A})$, this gives an exact category structure on $\mathcal{C}_i(\mathcal{A})$ inductively.

Definition 5.29 (The Category $\mathcal{P}_{\mathcal{A},i}$)
Let $\mathcal{A}$ be a split exact, additive category. For each integer $i \geq 0$, we define

$$\mathcal{P}_{\mathcal{A},i} := \mathcal{P}_{\mathcal{C}_i(\mathcal{A})} \tag{5.2.3}$$

as in Definition 5.13. In particular, we write:

$$\mathcal{P}_{R,i} := \mathcal{P}_{\mathcal{C}_i\left(\mathcal{F}\mathrm{ree}_R^{\mathrm{fg}}\right)} \tag{5.2.4}$$

for a ring R.

Pedersen-Weibel proved the following:

Theorem 5.30 ([PW85, Theorem B on page 167])
Let $\mathcal{A}$ be a split exact, additive category.

For each $i \geq 0$, we define

$$\left(\mathbb{K}_{\mathcal{A}}^{\mathrm{PW}}\right)_i := B\mathcal{P}_{\mathcal{A},i}. \tag{5.2.5}$$

Then for each $i \geq 0$, we have the delooping

$$\left(\mathbb{K}_{\mathcal{A}}^{\mathrm{PW}}\right)_i \simeq \Omega\left[\left(\mathbb{K}_{\mathcal{A}}^{\mathrm{PW}}\right)_{i+1}\right] \tag{5.2.6}$$

for all $i \geq 0$. Thus the sequence $\left\{\left(\mathbb{K}_{\mathcal{A}}^{\mathrm{PW}}\right)_i\right\}_{i=0}^{\infty}$ of spaces forms an Ω-spectrum.

Theorem 5.31 (The Non-connective Pedersen-Weibel K-theory Spectrum of Rings, [PW85, Theorem A on page 166])
Let R be a ring.

The spectrum

$$\mathbb{K}_R^{\mathrm{PW}} := \mathbb{K}^{\mathrm{PW}}_{\mathfrak{Free}_R^{\mathrm{fg}}} \tag{5.2.7}$$

is an Ω-spectrum, the so-called **non-connective Pedersen-Weibel algebraic K-theory spectrum of** R, whose homotopy groups are

$$\pi_i\left(\mathbb{K}_R^{\mathrm{PW}}\right) \cong \begin{cases} K_i(R) & \text{if } i \geq 0, \\ \begin{array}{c}\text{negative K-groups of } R \\ \text{defined by Bass in [Bas68]}\end{array} & \text{if } i < 0. \end{cases} \tag{5.2.8}$$

5.3. **The Universal Assembly.**

Definition 5.32 (Homotopy invariant, excisive, and strongly excisive functors) We say a functor

$$\mathbb{F} : \mathcal{S}\mathrm{paces} \to \mathcal{S}\mathrm{pectra}$$

is **homotopy invariant** if it takes homotopy equivlances to homotopy equivalences.

A homotopy invariant functor $\mathbb{F}$ is

(1) **excisive** if it preserves homotopy push-out squares, and $\mathbb{F}(\varnothing)$ is contractible;
(2) **strongly excisive** if it preserves arbitrary coproducts, up to homotopy equivalence.

Theorem 5.33 (Weiss-Williams/Universal Assembly, [WW95, Theorem 1.1. on page 333])
For any homotopy invariant functor

$$\mathbb{F} : \mathcal{S}\mathrm{paces} \to \mathcal{S}\mathrm{pectra},$$

there exists a strongly excisive functor

$$\mathbb{F}^{\%} : \mathcal{S}\mathrm{paces} \to \mathcal{S}\mathrm{pectra}$$

and a natural transformation

$$\alpha_{\mathrm{WW}} : \mathbb{F}^{\%} \Rightarrow \mathbb{F}, \tag{5.3.1}$$

the so-called **Weiss-Williams assembly**, or the **universal assembly**, such that the component

$$\alpha_{\mathrm{WW}} : \mathbb{F}^{\%}(\text{point}) \to \mathbb{F}(\text{point})$$

of the natural transformation is a homotopy equivalence. Moreover, $\mathbb{F}^{\%}$ and α_{WW} can be made to depend functorially on $\mathbb{F}$.

Proof. (Outline)

The point is that the functor

$$X \mapsto X_+ \wedge \mathbb{F}(\text{point}) \tag{5.3.2}$$

is a model for $\mathbb{F}^{\%}$; and the natural transformation

$$\alpha_{\mathrm{WW}} : \mathbb{F}^{\%} \Rightarrow \mathbb{F}$$

is induced by the constant map $X \to$ point. □

Theorem 5.34 ([WW95, Observation 1.3 on page 336])
If the homotopy invariant functor

$$\mathbb{F} : \mathcal{S}\mathsf{paces} \to \mathcal{S}\mathsf{pectra}$$

is already excisive, then the component

$$(\alpha_{\mathrm{WW}})_X : \mathbb{F}^{\%}(X) \to \mathbb{F}(X)$$

of the natural transformation

$$\alpha_{\mathrm{WW}} : \mathbb{F}^{\%} \Rightarrow \mathbb{F}$$

is a homotopy equialence for every compact X.
If $\mathbb{F}$ is strongly excisive, then $(\alpha_{\mathrm{WW}})_X$ is a homotopy equivalence for all X.

Theorem 5.35 (Universal Property of the Weiss-Williams Assembly, [WW95, page 336])
Suppose the homotopy invariant functor

$$\mathbb{F} : \mathcal{S}\mathsf{paces} \to \mathcal{S}\mathsf{pectra}$$

admits another natural transformation

$$\beta : \mathbb{E} \Rightarrow \mathbb{F}$$

from a strongly excisive functor $\mathbb{E}$. Then the diagram

$$\begin{tikzcd} \mathbb{E}^{\%} \arrow[r, "\alpha^{\mathbb{E}}_{\mathrm{WW}}"] \arrow[d, "\beta^{\%}"'] & \mathbb{E} \arrow[d, "\beta"] \\ \mathbb{F}^{\%} \arrow[r, "\alpha^{\mathbb{F}}_{\mathrm{WW}}"'] & \mathbb{F} \end{tikzcd} \tag{5.3.3}$$

commutes.

We conclude by the following remark. In Diagram (5.3.3), if the component

$$\beta_{\mathrm{point}} : \mathbb{E}(\mathrm{point}) \to \mathbb{F}(\mathrm{point})$$

of the natural transformation β is a homotopy equivalence, then

$$\beta^{\%}_X : \mathbb{E}^{\%}(X) \to \mathbb{F}^{\%}(X)$$

is a homotopy equivalence for all X by the Eilenberg-Steenrod Uniqueness Theorem ([ES52, Theorem 10.1 on page 100–101]). So β is homotopic to $\alpha^{\mathbb{F}}_{\mathrm{WW}}$ after identifying the source and target.

5.4. **Constructing the Universal Assembly.** We recall the notions of ringoids and modules over them as in [WW95, page 337–338].

Definition 5.36 (Ringoid, [WW95, page 337])
A **ringoid** is a small category in which
(1) all morphism sets come equipped with an Abelian group structure,
(2) composition of morphisms is bilinear.

Example 5.37 (The Ringoid $R[\mathcal{C}]$ of the Category $\mathcal{C}$ Over the Ring R). Let R be a ring, and $\mathcal{C}$ be a small category. We define the ringoid $R[\mathcal{C}]$ to be the category having the same objects as $\mathcal{C}$, and the morphism set

$$\mathrm{mor}_{R[\mathcal{C}]}(x, y) := R\langle \mathrm{mor}_{\mathcal{C}}(x, y)\rangle$$

is the free left R-module generated by the set $\mathrm{mor}_{\mathcal{C}}(x, y)$. When $\mathcal{C} = G$ is a group, considered as a category of one object, then the ringoid $R[\mathcal{C}]$ is the group ring $R[G]$, considered as a category of one object, hence justifying the notation.

The category $R[\mathcal{C}]$ is also referred as the **R-category associated to** $\mathcal{C}$ in [DL98, page 212].

Definition 5.38 (Modules over Ringoids, [WW95, page 338])
Let $\mathcal{R}$ be a ringoid.

A **left $\mathcal{R}$-module** is a functor

$$f : \mathcal{R} \to \mathcal{A}\mathrm{belian}$$

from $\mathcal{R}$ to the category of Abelian groups, such that the induced map

$$f : \mathrm{mor}_{\mathcal{R}}(x, y) \to \mathrm{mor}_{\mathcal{A}\mathrm{belian}}(f(x), f(y))$$

is a group homomorphism. A **right $\mathcal{R}$-module** is a left $\mathcal{R}^{op}$-module.

A left $\mathcal{R}$-module f is
(1) **free on one generator** if it is representable:

$$f(-) \cong \mathrm{mor}_{\mathcal{R}}(x, -)$$

for some $x \in \mathcal{R}$;
(2) **finitely generated free** if it is isomorphic to an arbitrary direct sum of representable ones;
(3) **projective** if it is a direct summand of a free one;
(4) **finitely generated projective** if it is a direct summand of a finitely generated free one.

The category $\mathcal{M}\mathrm{odule}_{\mathcal{R}}$ of left $\mathcal{R}$-modules and natural transformations forms an Abelian category. The subcategory $\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_{\mathcal{R}}$ is a split exact, additive category. We can then use the Pedersen-Weibel's construction as in Theorem 5.30 to get an Ω-spectrum

$$\mathbb{K}^{\mathrm{PW}}_{\mathcal{R}} := \mathbb{K}^{\mathrm{PW}}_{\mathcal{F}\mathrm{ree}^{\mathrm{fg}}_{\mathcal{R}}} \tag{5.4.1}$$

Now, when $\mathcal{R} = R$ is a ring, considered as a category of one object, there is a natural equivalence

$$\mathfrak{Free}^{\mathrm{fg}}_{\mathcal{R}} \simeq \mathfrak{Free}^{\mathrm{fg}}_{R}$$

of categories. In this case, the spectra $\mathbb{K}^{\mathrm{PW}}_{\mathcal{R}}$ and $\mathbb{K}^{\mathrm{PW}}_{R}$ have the same homotopy groups.

Let us construct a functor $\mathbb{Y} : \mathcal{S}\mathrm{paces} \to \mathcal{S}\mathrm{pectra}$. Let $\Pi(X)$ be the fundamental groupoid of the topological space X. Given a ring R, we can form the ringoid $R[\Pi(X)]$ as in Example 5.37. Our functor is then given by:

Definition 5.39 (The Functor $\mathbb{Y}$)
Let R be a ring. We define the functor $\mathbb{Y}$ by

$$\begin{aligned} \mathbb{Y} : \mathcal{S}\mathrm{paces} &\to \mathcal{S}\mathrm{pectra} \\ X &\mapsto \mathbb{K}^{\mathrm{PW}}_{R[\Pi(X)]} \end{aligned} \tag{5.4.2}$$

Clearly, the functor $\mathbb{Y}$ is homotopy invariant, so Theorem 5.34 asserts the existence of the universal assembly map

$$\alpha_{\mathrm{WW}} : \mathbb{Y}^{\%} \to \mathbb{Y}.$$

We will construct this assembly explicitly following [Spe04]. The key ingredient is an analogue of the Loday pairing for the Pedersen-Weibel spectrum.

Lemma 5.40 ([Spe04, Lemma 4.6 on page 16 and Lemma 4.7 on page 17])
Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be split exact, additive categories, considered as symmetric monoidal categories in the canonical way.

Suppose $\otimes : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$ is a pairing of symmetric monoidal categories. Then

(i) for all i, $j \geq 0$, there is an induced pairing

$$\otimes^i_j : \mathcal{C}_i(\mathcal{A}) \times \mathcal{C}_j(\mathcal{B}) \to \mathcal{C}_{i+j}(\mathcal{C}) \tag{5.4.3}$$

of symmetric monoidal categories.

(ii) Moreover, the collection $\{\otimes\}^{\infty}_{i,j=0}$ of pairings assemble to give a pairing

$$\widehat{\otimes} : \mathbb{K}^{\mathrm{PW}}_{\mathcal{A}} \wedge \mathbb{K}^{\mathrm{PW}}_{\mathcal{B}} \to \mathbb{K}^{\mathrm{PW}}_{\mathcal{C}}$$

of spectra.

Let us apply this result to ringoids.

Definition 5.41 (Tensor Product of Ringoids)
Let $\mathcal{R}$, $\mathcal{S}$ be ringoids. The tensor product of $\mathcal{R}$ and $\mathcal{S}$ is the ringoid $\mathcal{R} \otimes \mathcal{S}$ whose objects are given by pairs (r, s) of objects in $\mathcal{R}$ and $\mathcal{S}$, and the collection of morphisms is given by

$$\mathrm{Hom}_{\mathcal{R}\otimes\mathcal{S}}\left((r_1, s_1), (r_2, s_2)\right) := \mathrm{Hom}_{\mathcal{R}}\left(r_1, r_2\right) \otimes_{\mathbb{Z}} \mathrm{Hom}_{\mathcal{S}}\left(s_1, s_2\right). \tag{5.4.4}$$

Composition is given by

$$(f_1 \otimes f_2) \circ (g_1 \otimes g_2) := (f_1 \circ g_1) \otimes (f_2 \circ g_2), \tag{5.4.5}$$

and then extended linearly.

When the ringoids are rings considered as categories with one object, this tensor product becomes tensor product of rings in the usual sense. Furthermore, the tensor product of ringoids extends to a tensor product of modules over ringoids as defined in Definition 5.38.

Example 5.42 . Let R be a ring, and $\mathcal{C}$ be a small category. We can then form the ringoids $\mathbb{Z}[\mathcal{C}]$, $R[\mathcal{C}]$ as in Example 5.37. Tensor product of ringoids then gives

$$R \otimes \mathbb{Z}[\mathcal{C}] \cong R[\mathcal{C}]. \tag{5.4.6}$$

Moreover, it also gives a pairing

$$\otimes : \mathfrak{Free}_R^{\mathrm{fg}} \times \mathfrak{Free}_{\mathbb{Z}[\mathcal{C}]}^{\mathrm{fg}} \to \mathfrak{Free}_{R[\mathcal{C}]}^{\mathrm{fg}} \tag{5.4.7}$$

of symmetric monoidal categories as in Definition 5.15. Note that in this case, we are considering modules over ringoids as in Definition 5.38.

We are now ready to construct the universal assembly for the functor $\mathbb{Y}$. Let us recall from the proof of Theorem 5.33 that the functor

$$X \mapsto X_+ \wedge \mathbb{Y}\,(\text{point})$$

is a model for the functor $\mathbb{Y}^{\%} : \mathcal{S}\mathrm{paces} \to \mathcal{S}\mathrm{pectra}$.

Definition 5.43 (The Assembly $\widehat{\alpha}$)
Write $\Pi(X)$ to be the fundamental groupoid for the topological space X. Let R be a ring.

Define a map of spectra

$$\widehat{\alpha}_X : X_+ \wedge \mathbb{K}_R^{\mathrm{PW}} \to \mathbb{K}_{R[\Pi(X)]}^{\mathrm{PW}}$$

by having components

$$\begin{tikzcd}
X_+ \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[r, "c_+ \wedge \mathrm{id}"] \arrow[drr, dashed, "(\widehat{\alpha}_X)_n"'] & B\Pi(X)_+ \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[r, "J \wedge \mathrm{id}"] & \left(\mathbb{K}_{\mathbb{Z}[\Pi(X)]}^{\mathrm{PW}}\right)_{0+} \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[d, "\widehat{\otimes}"] \\
 & & \left(\mathbb{K}_{R[\Pi(X)]}^{\mathrm{PW}}\right)_n .
\end{tikzcd} \tag{5.4.8}$$

Here,

(i) the map $c : X \to B\Pi(X)$ is the classifying map for the universal cover of X;
(ii) the map $J : B\Pi(X)_+ \to \left(\mathbb{K}_{\mathbb{Z}[\Pi(X)]}^{\mathrm{PW}}\right)_{0+}$ is induced by the inclusion functor

$$\Pi(X) \to \mathrm{iso}\left(\mathbb{Z}[\Pi(X)]\right);$$

(iii) and the map $\widehat{\oplus}$ is the pairing map constructed in Lemma 5.40. In particular, it is induced by the tensor product of ringoids as constructed in Definition 5.41.

The collection of maps $\{\widehat{\alpha}_X \mid X \in \mathcal{S}\text{paces}\}$ then gives a natural transformation

$$\widehat{\alpha} : \mathbb{Y}^{\%} \Rightarrow \mathbb{Y} \tag{5.4.9}$$

for the functor $\mathbb{Y} : \mathcal{S}\text{paces} \to \mathcal{S}\text{pectra}$ constructed in Definition 5.39.

Theorem 5.44 ([HP04, Theorem 4.3 on page 39], [Spe04, Theorem 4.9 on page 18])
The natural transformation $\widehat{\alpha} : \mathbb{Y}^{\%} \Rightarrow \mathbb{Y}$ in Definition 5.43 is (homotopic to) the universal assembly map in the sense of Theorem 5.33.

5.5. **The Comparison.** We mimic Definition 3.1 to create two Loday-like assemblies.

Definition 5.45 (The Assembly Maps α_{free} and α_{proj})
Let R be a ring and G be a group.

We define two maps of spectra

$$BG \wedge \mathbb{K}_R^{\text{free}} \to \mathbb{K}_{R[G]}^{\text{free}} \tag{5.5.1}$$

$$BG \wedge \mathbb{K}_R^{\text{proj}} \to \mathbb{K}_{R[G]}^{\text{proj}} \tag{5.5.2}$$

by having components

$$\begin{tikzcd} BG \wedge \left(\mathbb{K}_R^{\text{free}}\right)_n \arrow[r, "j^{\text{free}} \wedge \mathrm{id}"] \arrow[dr, dashed, "(\alpha_{\text{free}})_n"'] & \left(\mathbb{K}_{\mathbb{Z}[G]}^{\text{free}}\right)_0 \wedge \left(\mathbb{K}_R^{\text{free}}\right)_n \arrow[d, "\gamma_{\text{free}}"] \\ & \left(\mathbb{K}_{R[G]}^{\text{free}}\right)_n, \end{tikzcd} \quad \begin{tikzcd} BG \wedge \left(\mathbb{K}_R^{\text{proj}}\right)_n \arrow[r, "j^{+} \wedge \mathrm{id}"] \arrow[dr, dashed, "(\alpha_{\text{proj}})_n"'] & \left(\mathbb{K}_{\mathbb{Z}[G]}^{\text{proj}}\right)_0 \wedge \left(\mathbb{K}_R^{\text{proj}}\right)_n \arrow[d, "\gamma_{\text{proj}}"] \\ & \left(\mathbb{K}_{R[G]}^{\text{proj}}\right)_n \end{tikzcd} \tag{5.5.3}$$

respectively. Here, the map j^+ is defined in Equation (3.1.3), and the map j^{free} is defined analogously. We then use the construction in Definition 3.3 to extend these two maps and get:

$$\alpha_{\text{free}} : BG_+ \wedge \mathbb{K}_R^{\text{free}} \to \mathbb{K}_{R[G]}^{\text{free}}, \tag{5.5.4}$$

$$\alpha_{\text{proj}} : BG_+ \wedge \mathbb{K}_R^{\text{proj}} \to \mathbb{K}_{R[G]}^{\text{proj}}. \tag{5.5.5}$$

In light of their definitions and Proposition 3.2, the maps α_{free} and α_{proj} are maps of spectra, well-defined up to weak-homotopy. The following result is an immediate consequence of the definitions.

Proposition 5.46
Let R be a ring satisfying the IBP, and G be a group.

The canonical map

$$\mathcal{S}_R \to \mathcal{P}_R$$

given in Equation (5.1.19) gives a weak equivalence

$$\mu : \mathbb{K}_R^{\text{free}} \to \mathbb{K}_R^{\text{proj}} \tag{5.5.6}$$

of spectra. Moreover, this weak equivalence fits into the following diagram

$$\begin{tikzcd} BG_+ \wedge \mathbb{K}_R^{\text{free}} \arrow[r, "\alpha_{\text{free}}"] \arrow[d, "\mu"'] & \mathbb{K}_{R[G]}^{\text{free}} \arrow[d, "\mu"] \\ BG_+ \wedge \mathbb{K}_R^{\text{proj}} \arrow[r, "\alpha_{\text{proj}}"'] & \mathbb{K}_{R[G]}^{\text{proj}} \end{tikzcd} \tag{5.5.7}$$

of spectra, which commutes up to weak homotopy.

The proof of the following result was outlined in [HP04], and later completed in [Spe04].

Theorem 5.47
Let R be a ring satisfying the IBP.
For each $i \geq 0$, there is a functor

$$G_i : \mathcal{C}_i\left(\mathfrak{Free}_R^{\text{fg}}\right) \to \mathfrak{Free}_{\Sigma^i R}^{\text{fg}}, \tag{5.5.8}$$

such that

(a) the collection $\{G_i\}_{i=0}^{\infty}$ of functors gives a homotopy equivalence

$$\mathfrak{g} : \mathbb{K}_R^{\text{PW}} \to \mathbb{K}_R^{\text{proj}} \tag{5.5.9}$$

of spectra [Spe04, Theorem 5.14 on page 26 and Proposition 5.15 on page 28].

(b) Moreover, the map $\mathfrak{g}$ is compatible with the pairings of spectra. More precisely, for each pair of non-negative integers m, n, there is a diagram

$$\begin{tikzcd} \left(\mathbb{K}_R^{\text{PW}}\right)_m \wedge \left(\mathbb{K}_R^{\text{PW}}\right)_n \arrow[r, "\widehat{\otimes}"] \arrow[d, "\mathfrak{g}_m \wedge \mathfrak{g}_n"'] & \left(\mathbb{K}_R^{\text{PW}}\right)_{m+n} \arrow[d, "\mathfrak{g}_{m+n}"] \\ \left(\mathbb{K}_R^{\text{proj}}\right)_m \wedge \left(\mathbb{K}_R^{\text{proj}}\right)_n \arrow[r, "\gamma_{\text{proj}}"'] & \left(\mathbb{K}_R^{\text{proj}}\right)_{m+n} \end{tikzcd} \tag{5.5.10}$$

commutes up to weak homotopy [Spe04, Corollary 6.3 on page 40].

Proof. (Outline) We outline the construction of the functors G_n here since this theorem is the main result of [Spe04]. We begin by constructing the functors

$$G_n : \mathcal{C}_n\left(\mathfrak{Free}_R^{\text{fg}}\right) \to \mathfrak{Free}_{\Sigma^n R}^{\text{fg}}$$

inductively.

Step 1: When $n = 0$, we have

$$\mathcal{C}_0\left(\mathfrak{Free}_R^{\text{fg}}\right) = \mathfrak{Free}_R^{\text{fg}}.$$

Thus, we define

$$G_0 := \mathrm{id}_{\mathfrak{Free}_R^{\mathrm{fg}}}\,. \tag{5.5.11}$$

Step 2: Secondly, let $\mathfrak{Free}_R^{\mathbb{N}}$ be the category of countably generated free R-modules and locally finite matrices over R. It is proven that the categories

$$\frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}} \simeq \mathfrak{Free}_{\Sigma R}^{\mathrm{fg}}$$

are equivalent [PW89, Proposition 6.1 on page 359].

The idea is as follows: choose an infinitely based R-module R^∞ in $\mathfrak{Free}_R^{\mathbb{N}}$, and observe that

$$\mathrm{End}_{\mathfrak{Free}_R^{\mathbb{N}}}\left(R^\infty\right) \cong \mathrm{cone}\left(R\right).$$

Now, the category $\mathfrak{Free}_R^{\mathbb{N}}$ is Karoubi-filtered by its subcategory $\mathfrak{Free}_R^{\mathrm{fg}}$, and the completely continuous endomorphisms of R^∞ form the ideal $m(R)$ of $\mathrm{cone}\,(R)$. Thus, we have

$$\mathrm{End}_{\frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}}}\left(R^\infty\right) \cong \Sigma R.$$

The canonical additive functor

$$\begin{aligned} \mathfrak{Free}_{\Sigma R}^{\mathrm{fg}} &\to \frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}} \\ \Sigma R &\mapsto R^\infty \end{aligned} \tag{5.5.12}$$

is therefore full and faithful. But every object of $\frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}}$ is either isomorphic to the zero module or to R^∞, so this functor is also an equivalence.

We denote by

$$\alpha : \frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}} \to \mathfrak{Free}_{\Sigma R}^{\mathrm{fg}} \tag{5.5.13}$$

the inverse equivalence functor of Equation (5.5.12).

Step 3: Thirdly, we define the functor

$$\begin{aligned} \beta : \mathcal{C}_+\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) &\to \mathfrak{Free}_R^{\mathbb{N}} \\ \{A_i\}_{i=0}^{\infty} &\mapsto \bigoplus_{i=0}^{\infty} A_i, \end{aligned}$$

for which $\mathcal{C}_+\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)$ is the subcategory of $\mathcal{C}_1\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)$ consisting objects $\{A_i\}_{i=-\infty}^{\infty}$ with $A_i = 0$ whenever $i < 0$.

Note that each A_i is finitely generated, so the direct sum $\bigoplus_{i=0}^{\infty} A_i$ is an object in $\mathfrak{Free}_R^{\mathbb{N}}$. Moreover, this functor induces a functor

$$\widehat{\beta} : \frac{\mathcal{C}_+\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)}{\mathfrak{Free}_R^{\mathrm{fg}}} \to \frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}}. \tag{5.5.14}$$

We comment that $\mathfrak{Free}_R^{\mathrm{fg}}$ is regarded as a subcategory of $\mathcal{C}_+\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)$ via the embedding

$$A \mapsto (A, 0, 0, \cdots).$$

Step 4: Next, for a filtered additive category $\mathcal{A}$ in the sense of Definition 5.26, we define the functor

$$\begin{aligned} \tau : \mathcal{C}_1(\mathcal{A}) &\to \frac{\mathcal{C}_+(\mathcal{A})}{\mathcal{A}} \\ \{A_i\}_{i=-\infty}^{\infty} &\mapsto \{A_i\}_{i=0}^{\infty}. \end{aligned} \tag{5.5.15}$$

Recall that a morphism ϕ in $\mathcal{C}_1(\mathcal{A})$ is an infinite matrix $(\phi_{i,j})_{i,j=-\infty}^{\infty}$. We define $\tau(\phi)$ to be the sub-matrix $(\phi_{i,j})_{i,j=1}^{\infty}$.

Step 5: We now define the functor

$$G_n : \mathcal{C}_n\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \to \mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}$$

for $n \geq 1$ inductively. When $n = 1$, we define G_1 to be the composition

$$\begin{tikzcd} \mathcal{C}_1\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \arrow[r, "\tau"] \arrow[rrr, bend right, dashed, "G_1"'] & \frac{\mathcal{C}_+\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)}{\mathfrak{Free}_R^{\mathrm{fg}}} \arrow[r, "\widehat{\beta}"] & \frac{\mathfrak{Free}_R^{\mathbb{N}}}{\mathfrak{Free}_R^{\mathrm{fg}}} \arrow[r, "\alpha"] & \mathfrak{Free}_{\Sigma R}^{\mathrm{fg}}. \end{tikzcd} \tag{5.5.16}$$

Now suppose the functor

$$G_n : \mathcal{C}_n\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \to \mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}$$

is defined. As pointed out in Example 5.28, the categories $\mathcal{C}_i\left(\mathcal{C}_j\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)\right)$ and $\mathcal{C}_{i+j}\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)$ are isomorphic. We define the functor G_{n+1} to be the composition:

$$\begin{tikzcd}
\mathcal{C}_{n+1}\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \arrow[r, "\cong"] \arrow[drrrr, dashed, "G_{n+1}"'] & \mathcal{C}_1\left(\mathcal{C}_n\left(\mathfrak{Free}_R^{\mathrm{fg}}\right)\right) \arrow[r, "\mathcal{C}_1(G_n)"] & \mathcal{C}_1\left(\mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}\right) \arrow[r, "\tau"] & \dfrac{\mathcal{C}_+\left(\mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}\right)}{\mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}} \arrow[d, "\widehat{\beta}"] \\
& & & \dfrac{\mathfrak{Free}_{\Sigma^n R}^{\mathbb{N}}}{\mathfrak{Free}_{\Sigma^n R}^{\mathrm{fg}}} \arrow[d, "\alpha"] \\
& & & \mathfrak{Free}_{\Sigma^{n+1} R}^{\mathrm{fg}}.
\end{tikzcd} \tag{5.5.17}$$

This completes the construction of the functors

$$G_i : \mathcal{C}_i\left(\mathfrak{Free}_R^{\mathrm{fg}}\right) \to \mathfrak{Free}_{\Sigma^i R}^{\mathrm{fg}},$$

and we refer readers to [Spe04, Theorem 5.14 on page 26, Proposition 5.15 on page 28 and Corollary 6.3 on page 40] for checking the remaining properties. $\square$

A consequence of Theorem 5.47 is the following corollary:

Corollary 5.48
Let R be a ring satisfying IBP, and G be a group.

Consider G as a groupoid in the canonical way, then the component

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\mathrm{PW}} \to \mathbb{K}_{R[G]}^{\mathrm{PW}} \tag{5.5.18}$$

of the universal assembly map $\widehat{\alpha}$, as constructed in Definition 5.43, is weakly homotopic to the assembly map

$$\alpha_{\mathrm{proj}} : BG_+ \wedge \mathbb{K}_R^{\mathrm{proj}} \to \mathbb{K}_{R[G]}^{\mathrm{proj}} \tag{5.5.19}$$

constructed in Definition 5.45.

Proof. We need to check the diagram

$$\begin{tikzcd}
BG_+ \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[r, "c_+ \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \mathfrak{g}_n"'] & BG_+ \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[r, "J \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \mathfrak{g}_n"'] & \left(\mathbb{K}_{\mathbb{Z}[G]}^{\mathrm{PW}}\right)_0 \wedge \left(\mathbb{K}_R^{\mathrm{PW}}\right)_n \arrow[r, "\widehat{\otimes}"] \arrow[d, "\mathrm{id} \wedge \mathfrak{g}_n"'] & \left(\mathbb{K}_{R[G]}^{\mathrm{PW}}\right)_n \arrow[d, "\mathfrak{g}_n"] \\
BG_+ \wedge \left(\mathbb{K}_R^{\mathrm{proj}}\right)_n \arrow[r, "c_+ \wedge \mathrm{id}"'] & BG_+ \wedge \left(\mathbb{K}_R^{\mathrm{proj}}\right)_n \arrow[r, "j^+ \wedge \mathrm{id}"'] & \left(\mathbb{K}_{\mathbb{Z}[G]}^{\mathrm{proj}}\right)_0 \wedge \left(\mathbb{K}_R^{\mathrm{proj}}\right)_n \arrow[r, "\gamma_{\mathrm{proj}}"'] & \left(\mathbb{K}_{R[G]}^{\mathrm{proj}}\right)_n
\end{tikzcd} \tag{5.5.20}$$

commutes up to weak homotopy for all $n \geq 0$. Here,

(1) the classifying maps

$$c_+ : BG_+ \to BG_+$$

for the universal cover of BG_+ is just the identity, so the left square commutes up to homotopy automatically.

(2) The middle square commutes up to homotopy by the definitions of J and j^+.

(3) The right square commutes up to weak homotopy by part (b) of Theorem 5.47.

Finally, the map $\mathfrak{g}_n$ is a homotopy equivalence by part (a) of Theorem 5.47. So the claim holds as desired. □

Corollary 5.49 (Corollary of Corollary 5.48)
Let R be a ring satisfying IBP, and G be a group.

Consider G as a groupoid in the canonical way, then the component

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\mathrm{PW}} \to \mathbb{K}_{R[G]}^{\mathrm{PW}} \tag{5.5.21}$$

of the universal assembly map $\widehat{\alpha}$, as constructed in Definition 5.43, is weakly homotopic to the assembly map

$$\alpha_{\mathrm{free}} : BG_+ \wedge \mathbb{K}_R^{\mathrm{free}} \to \mathbb{K}_{R[G]}^{\mathrm{free}} \tag{5.5.22}$$

constructed in Definition 5.45.

5.6. **An Explicit Formula for the Universal Assembly.** We now relate Theorem 3.8 to the universal assembly map.

Theorem 5.50
Let R be a ring and G be a group.

There is a subgroup $\mathrm{AHSS}\,(BG)_{1,i}^{\infty}$ of $\pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\mathrm{PW}}\right)$ which is a quotient of the E^2-term

$$\mathrm{AHSS}\,(BG)_{1,i}^{2} \cong G_{ab} \otimes K_i(R)$$

from the Atiyah-Hirzebruch spectral sequence of the classifying space BG of G with coefficients in the Pedersen-Weibel K-theory spectrum $\mathbb{K}_R^{\mathrm{PW}}$ of R.

For $i > 0$, the filler of the diagram

$$\begin{tikzcd}
\mathrm{AHSS}\,(BG)_{1,i}^{2} \cong G_{ab} \otimes K_i(R) \arrow[r, dashed] \arrow[d, two heads] & K_{i+1}\left(R[G]\right) \\
\mathrm{AHSS}\,(BG)_{1,i}^{\infty} \arrow[r, hook] & \pi_{i+1}\left(BG \wedge \mathbb{K}_R^{\mathrm{PW}}\right) \arrow[u, "\pi_{i+1}\left(\widehat{\alpha}_{BG}\right)"']
\end{tikzcd} \tag{5.6.1}$$

is induced by

$$\begin{aligned} G \times K_i(R) &\to K_{i+1}\left(R[G]\right) \\ (g, [f]) &\mapsto [f] \star \{\mu_g\}. \end{aligned} \tag{5.6.2}$$

(Note that AHSS $(BG)^{\infty}_{1,0} = 0$ by regularity of R.) This provides an explicit formula for the universal assembly map when restricted onto the subgroup AHSS $(BG)^{\infty}_{1,i}$.

Proof. From Corollary 5.49, we know the universal assembly $\widehat{\alpha}_{BG}$ induces the same map on homotopy groups as the assembly map α_{free}. So we need to verify the formula for α_{free}.

As pointed out in Equation (5.1.9), the pairing map γ_{free} used in constructing the assembly α_{free} is homotopic to the original Loday pairing γ_{Loday}. Therefore, by repeating the same proofs as in Proposition 3.6 and Proposition 3.7, we see that the bottom composition of

$$\begin{tikzcd}
\pi_{i+1}\left(\left(\bigvee_{g\in G}\mathbb{S}^1\right)\wedge\mathbb{K}_R^{\text{free}}\right) \arrow[r, "(\mathrm{i}_G)_*"] \arrow[d, phantom, "\wr\|"] & \pi_{i+1}\left(BG\wedge\mathbb{K}_R^{\text{free}}\right) \arrow[r, "\alpha_{\text{free}}"] \arrow[d, phantom, "\|"] & K_{i+1}\left(R[G]\right) \arrow[d, phantom, "\|"] \\
\bigoplus_{g\in G} K_i(R) \arrow[r] & \pi_{i+1}\left(BG\wedge\mathbb{K}_R^{\text{free}}\right) \arrow[r] & K_{i+1}\left(R[G]\right)
\end{tikzcd} \tag{5.6.3}$$

sends the element $[f] \in K_i(R)$ in the summand in $\bigoplus_{g\in G} K_i(R)$ labelled by $g \in G$ to the element

$$[f] \star \{\mu_g\} \in K_{i+1}\left(R[G]\right),$$

for which $\mu_g \in \mathrm{Aut}\left(R[G]\right)$ is multiplication by g. This completes the proof. □

The universal assembly then admits the following version of Corollary 4.1.

Corollary 5.51 (The Universal Assembly on π_2 of an Integral Group Ring) Let G be a group. The universal assembly map

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}^{\text{PW}}_{\mathbb{Z}} \to \mathbb{K}^{\text{PW}}_{\mathbb{Z}[G]} \tag{5.6.4}$$

for the integral group ring $\mathbb{Z}[G]$ on π_2, when restricted onto the subgroup

$$K_2\left(\mathbb{Z}\right) \oplus \left[G_{ab} \oplus K_1(\mathbb{Z})\right],$$

is given by the formula:

$$K_2(i) \oplus \widehat{\Phi}_2 : K_2\left(\mathbb{Z}\right) \oplus \left[G_{ab} \oplus K_1(\mathbb{Z})\right] \to K_2(\mathbb{Z}[G]), \tag{5.6.5}$$

for which $i : \mathbb{Z} \to \mathbb{Z}[G]$ is the inclusion, and $\widehat{\Phi}_2$ is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\to K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto -\{\pm 1, g\}_{\text{St}}. \end{aligned} \tag{5.6.6}$$

Proof. We have $-\{\pm 1, g\}_{\text{St}}$ instead of $\{\pm 1, g\}_{\text{St}}$ because of Proposition 2.4 (iii). □

Because of Question 5.23, it is unclear if Corollary 3.9 (or its variants) holds for the universal assembly. However, Waldhausen showed that when $R = \mathbb{Z}$, the assembly

$$\pi_1(\alpha_{\text{free}}) : \pi_1(BG_+ \wedge \mathbb{K}^{\text{free}}_{\mathbb{Z}}) \to K_1(\mathbb{Z}[G])$$

on π_1 is the usual map

$$\begin{aligned} \{1,-1\} \oplus G_{ab} &\to K_1(\mathbb{Z}[G]) \\ (\pm 1) \oplus g &\mapsto \{\pm \mu_g\}\,. \end{aligned}$$

See [Wal78b, Assertion 15.8 on page 229]. His proof involves rewriting the assembly map in terms of Quillen's Q-construction, and then verified the formula at the simplicial level. Therefore, it seems that, for a regular ring R, the universal assembly map on π_1 is given by the formula

$$K_1(i) \oplus \widehat{\Phi}_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \to K_1\left(R[G]\right), \tag{5.6.7}$$

where $i : R \to R[G]$ is the inclusion, and the map $\widehat{\Phi}_1$ is induced by the map

$$\begin{aligned} G \times K_0(R) &\to K_1(R[G]) \\ (g,[P]) &\mapsto \left\{ \begin{array}{cccc} \widehat{\mathfrak{h}}_g : & P \otimes_R R[G] & \to & P \otimes_R R[G] \\ & x \otimes u & \mapsto & x \otimes ug \end{array} \right\}. \end{aligned} \tag{5.6.8}$$

Again, we have $\left\{\widehat{\mathfrak{h}}_g\right\}$ instead of $\{\mathfrak{h}_g\}$ as in Corollary 3.9 because of Proposition 2.7. But a rigorous proof is not known to the author.

5.7. **A Final Remark on Extending the Loday Pairing.** One might ask why we use the Gersten-Wagoner delooping to extend the Loday pairing

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

to

$$\gamma'_{\text{Loday}} : \left[K_0(R) \times BGL(R)^+\right] \wedge \left[K_0(S) \times BGL(S)^+\right] \to K_0(R \otimes S) \times BGL(R \otimes S)^+\,,$$

instead of using the "obvious approach":

$$\begin{aligned} \gamma_{\text{naive}} : \left[K_0(R) \times BGL(R)^+\right] \wedge \left[K_0(S) \times BGL(S)^+\right] &\to K_0(R \otimes S) \times BGL(R \otimes S)^+ \\ ([P],x) \wedge ([Q],y) &\mapsto ([P \otimes Q], \gamma_{\text{Loday}}(x,y))\,. \end{aligned} \tag{5.7.1}$$

This approach is fine if one only wants to construct a pairing map. However, if we look at the induced product map

$$\begin{aligned} \star_{\text{naive}} : K_i(R) \otimes K_j(S) &\to K_{i+j}(R \otimes S) \\ [f] \otimes [g] &\mapsto [\gamma_{\text{naive}} \circ (f \wedge g)]\,, \end{aligned} \tag{5.7.2}$$

we immediately see that it is the zero map when $i = 0$. If this naive pairing extends to a (weak) pairing

$$\gamma_{\text{naive}} : \mathbb{K}_R^{\text{GW}} \wedge \mathbb{K}_S^{\text{GW}} \to \mathbb{K}_{R\otimes S}^{\text{GW}} \tag{5.7.3}$$

of spectra, then one can define an assembly map

$$\alpha_{\text{naive}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}} \tag{5.7.4}$$

by mimicking Definition 3.3. However, the cokernel $\operatorname{coker}(\pi_1(\alpha_{\mathrm{naive}}))$ of this assembly on π_1 will not be isomorphic to the classical Whitehead group $\mathrm{Wh}_1(G)$ when $R = \mathbb{Z}$. Since everyone agrees that the cokernel of the universal assembly on π_1 should be isomorphic to $\mathrm{Wh}_1(G)$, the assembly α_{naive} is not the correct one, so does the pairing γ_{naive}.

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